

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

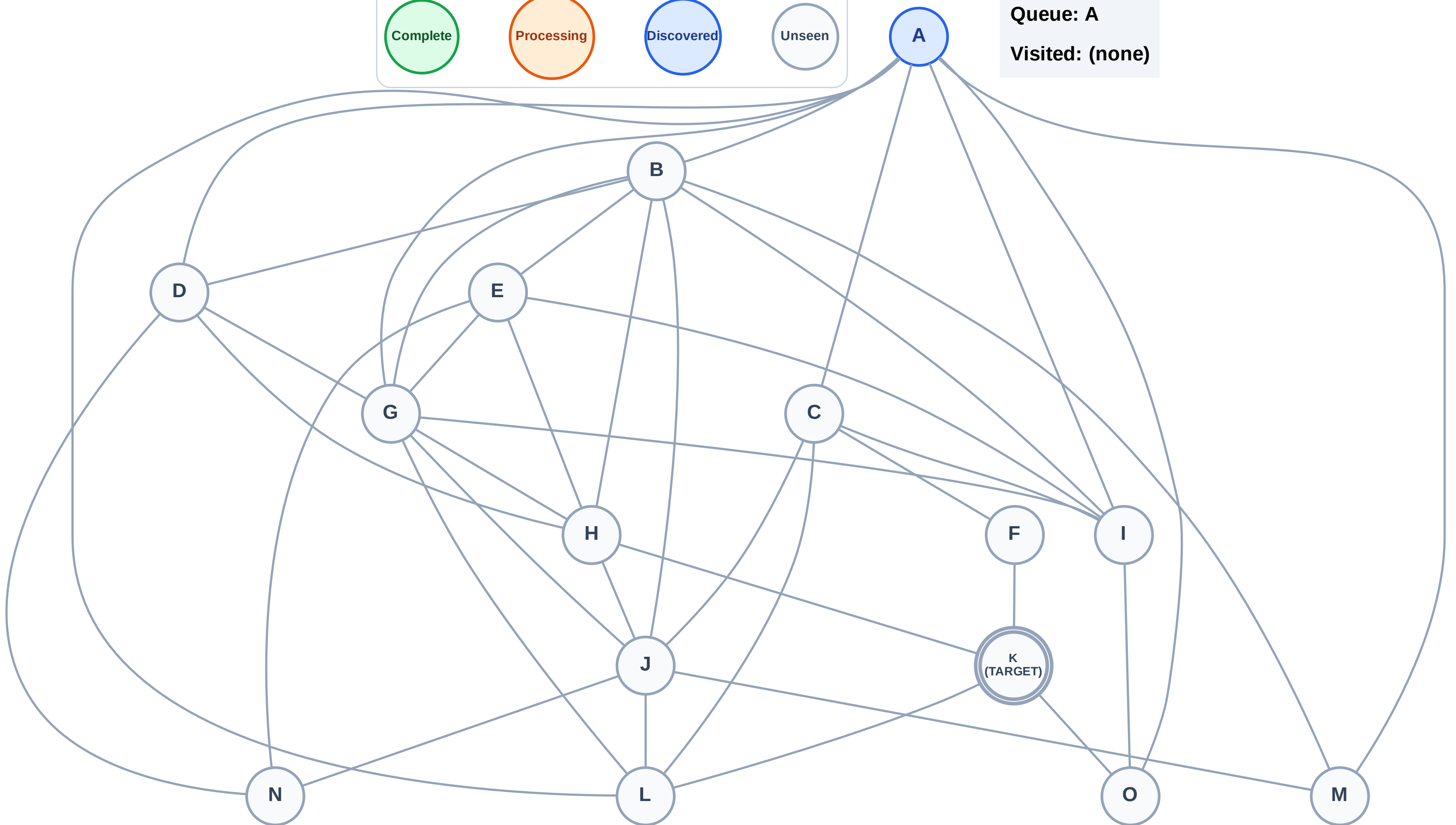


Discovered



Unseen

Queue: A
Visited: (none)



BFS Traversal

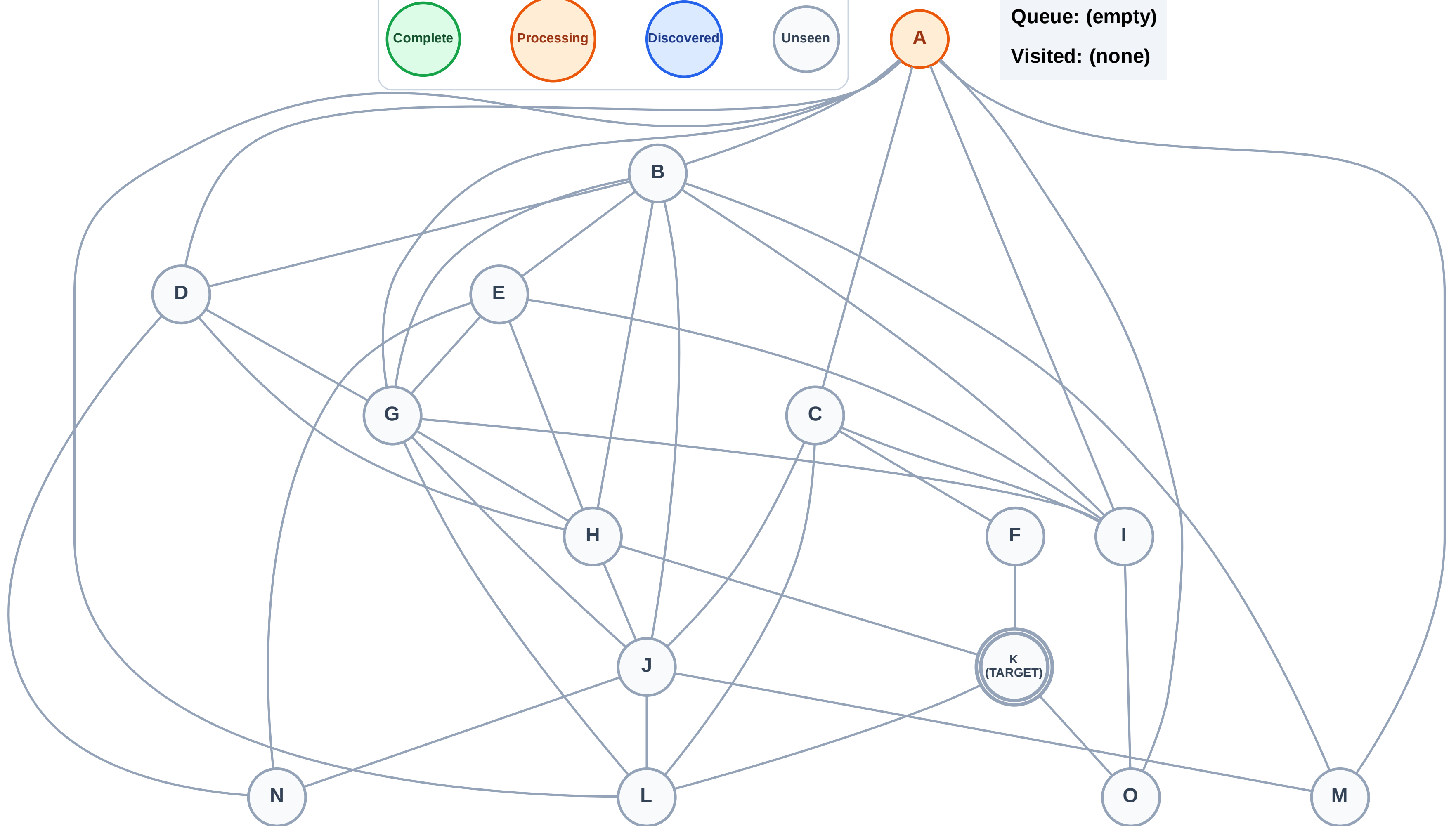
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



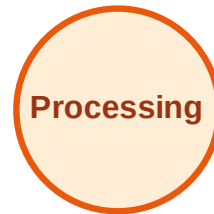
BFS Traversal

Step 3: Discover B, C, D, G, I, L, M, O from A

Legend



Complete



Processing



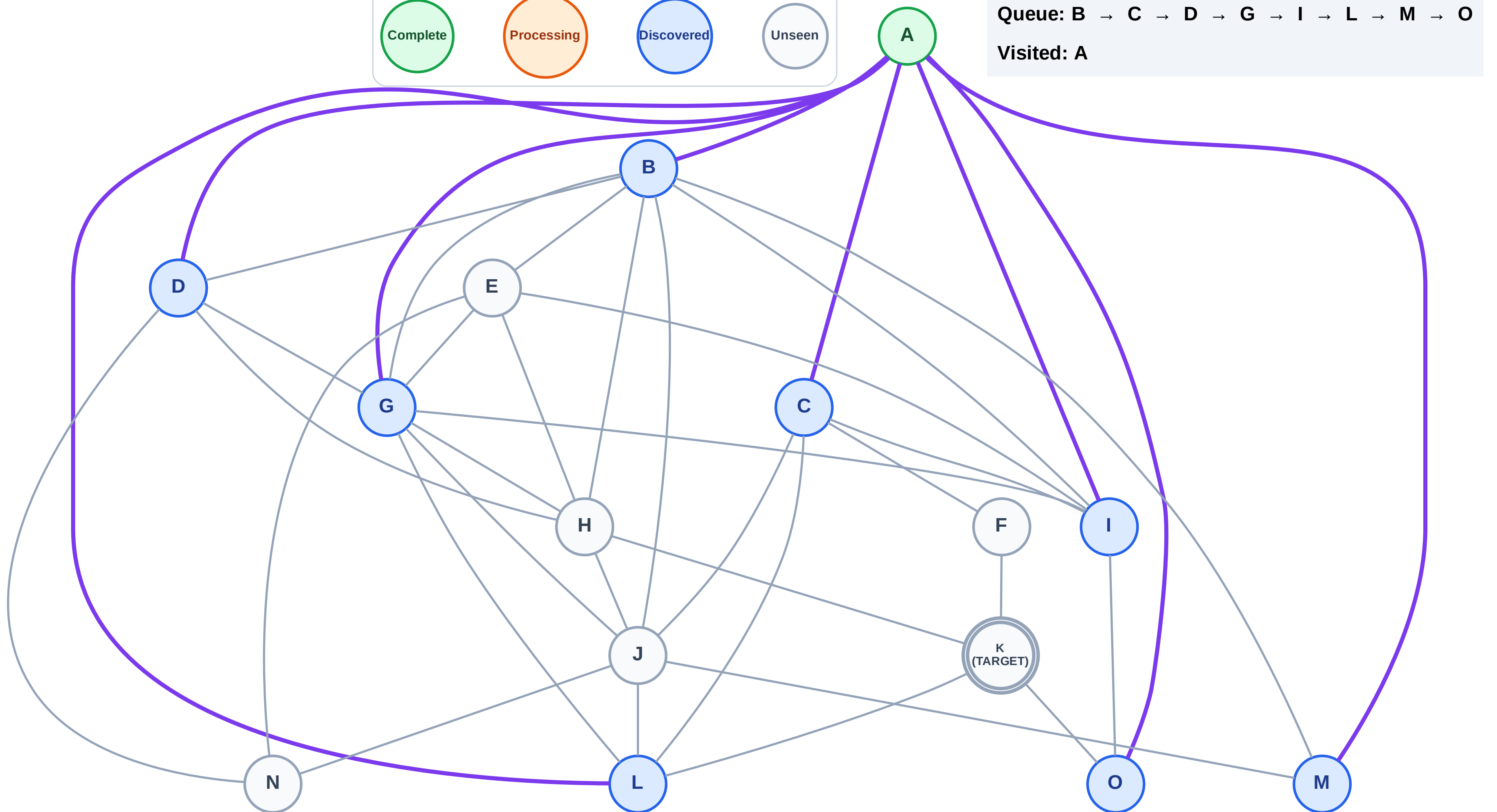
Discovered



Unseen

Queue: B → C → D → G → I → L → M → O

Visited: A



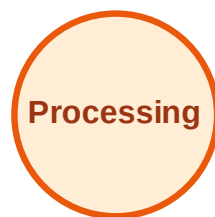
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



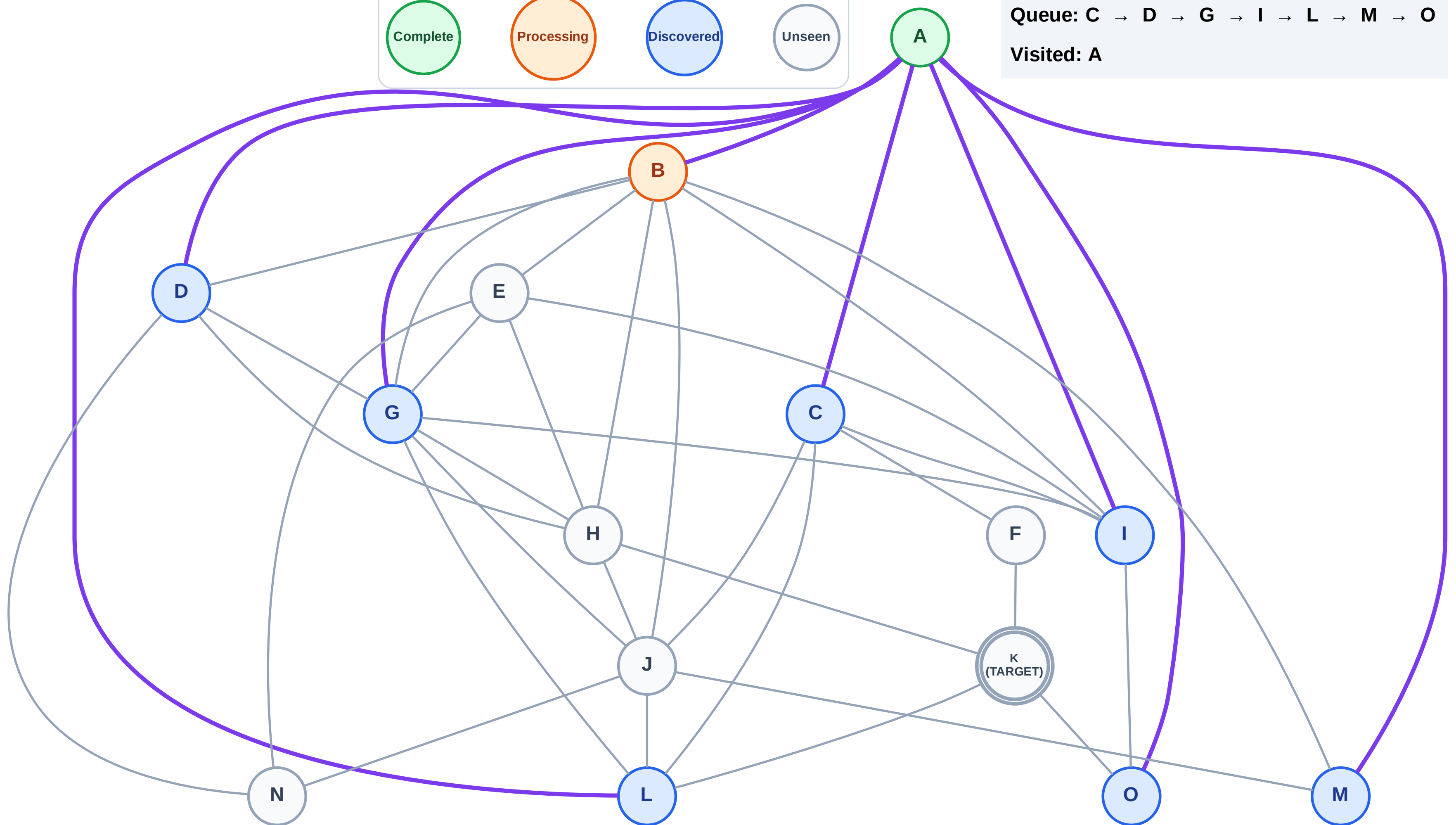
Discovered



Unseen

Queue: C → D → G → I → L → M → O

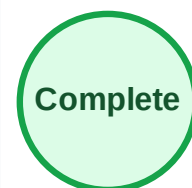
Visited: A



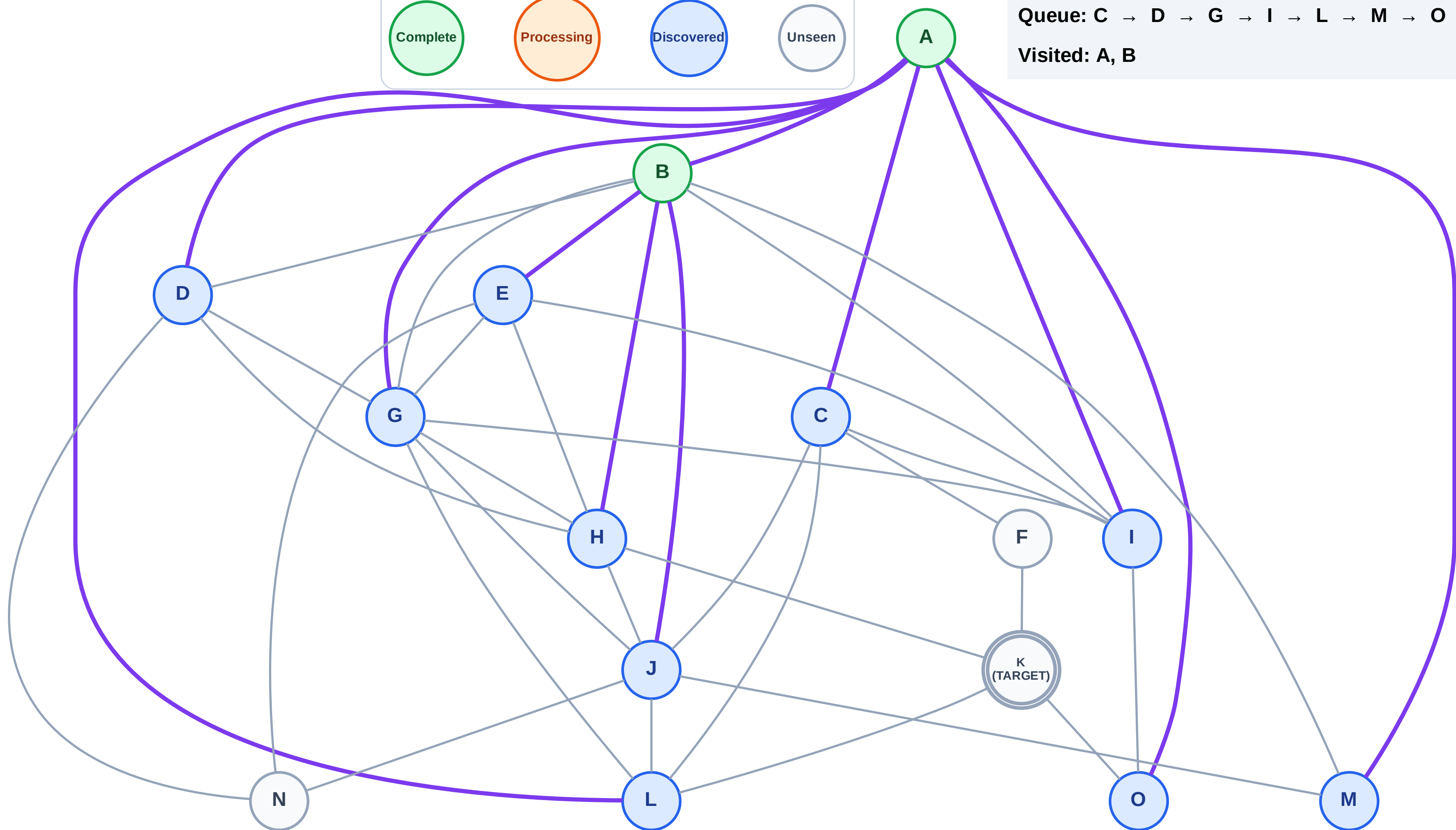
BFS Traversal

Step 5: Discover E, H, J from B

Legend



Queue: C → D → G → I → L → M → O → E → H → J
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

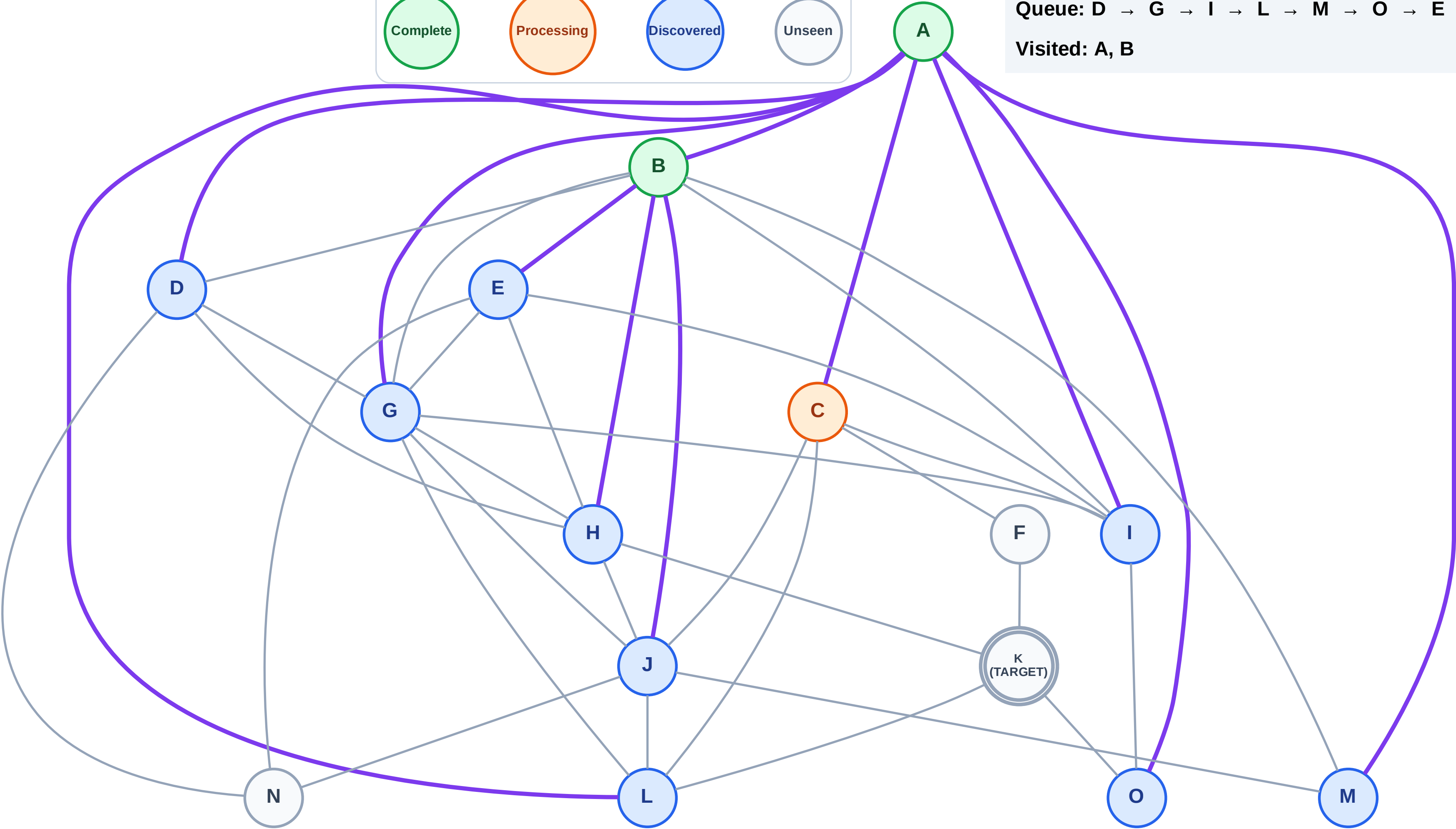
Processing

Discovered

Unseen

Queue: D → G → I → L → M → O → E → H → J

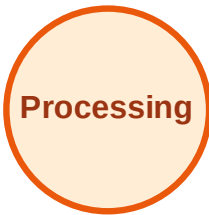
Visited: A, B



BFS Traversal

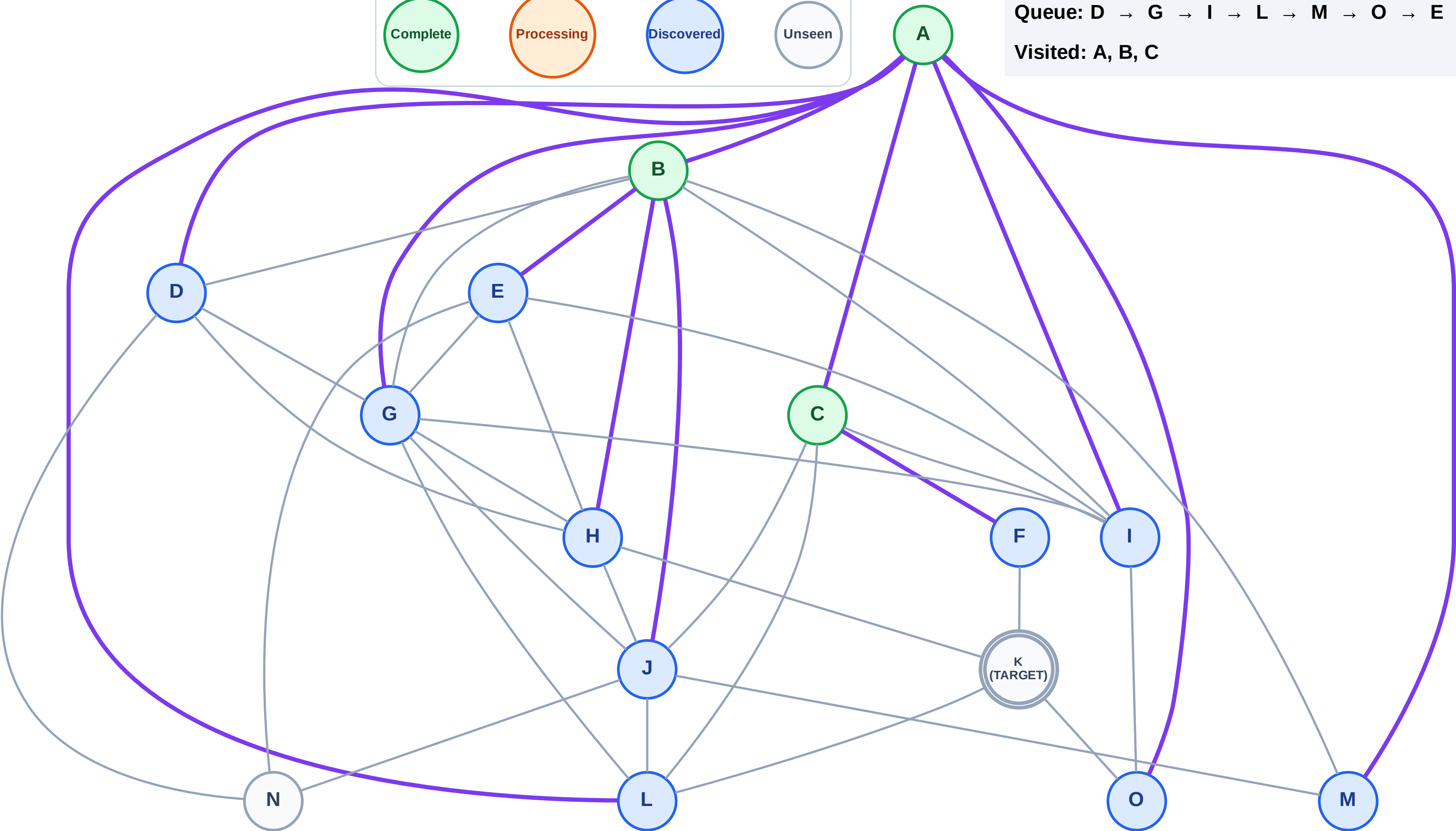
Step 7: Discover F from C

Legend



Queue: D → G → I → L → M → O → E → H → J → F

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

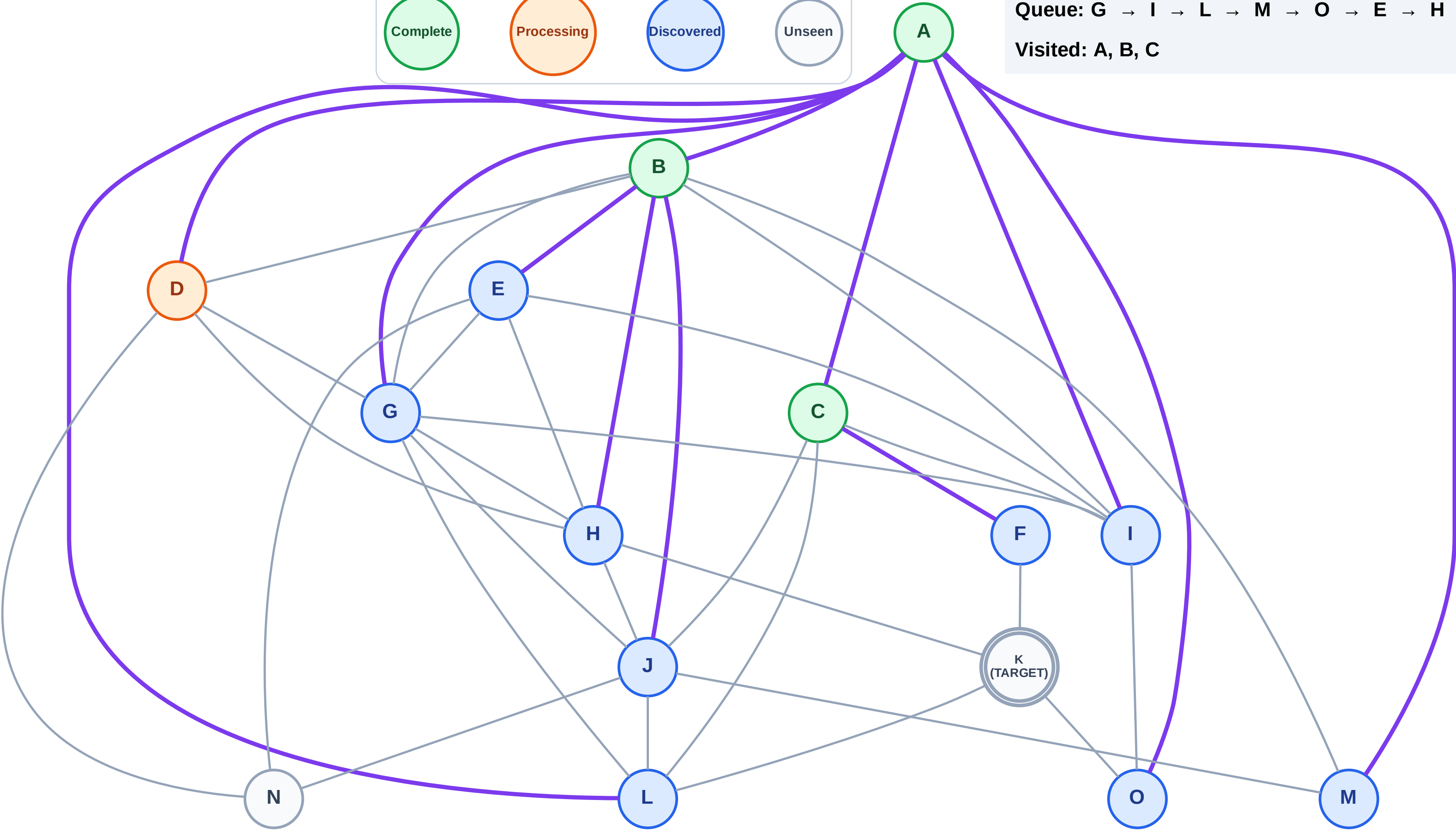
Processing

Discovered

Unseen

Queue: G → I → L → M → O → E → H → J → F

Visited: A, B, C



BFS Traversal

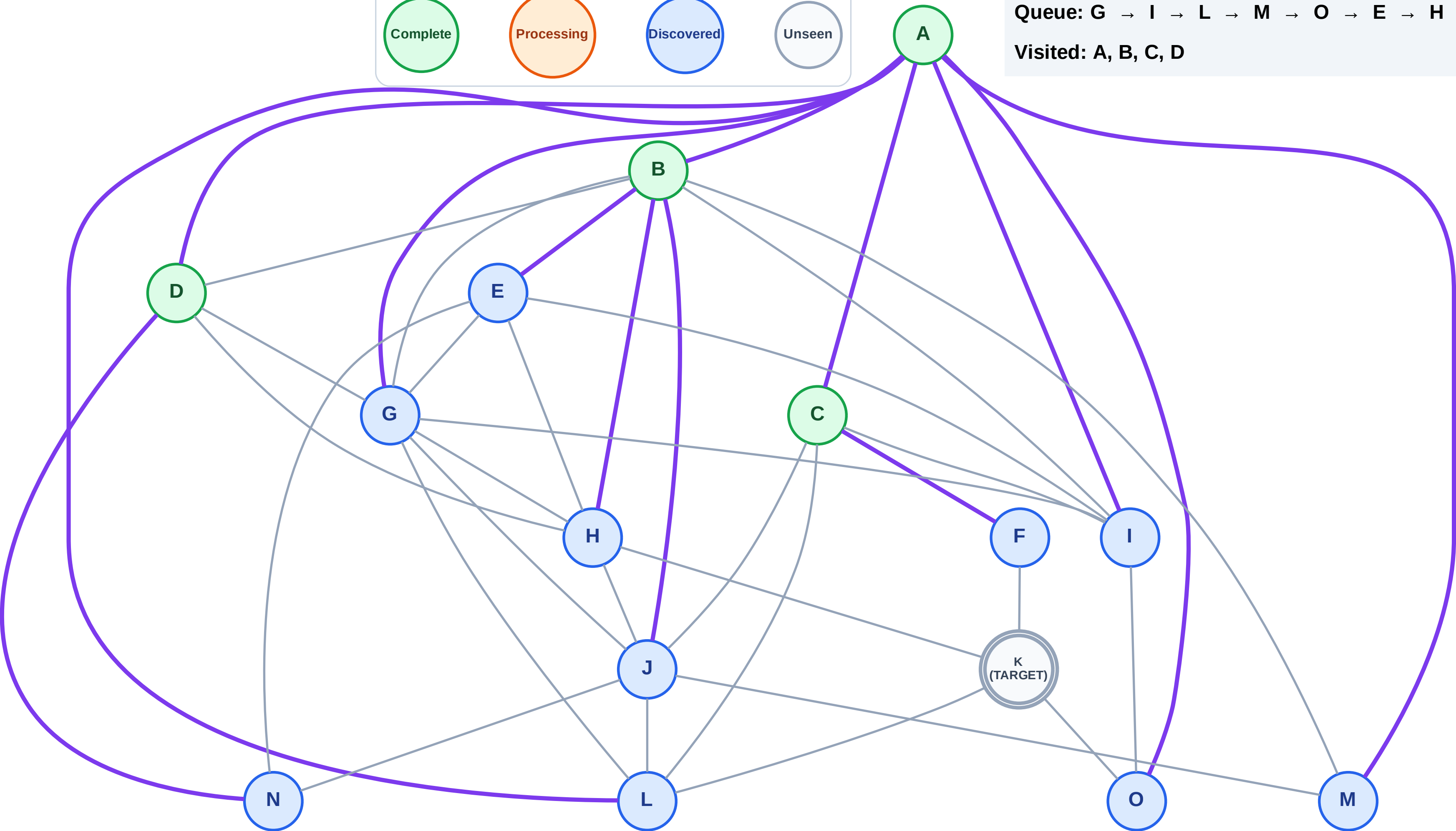
Step 9: Discover N from D

Legend



Queue: G → I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



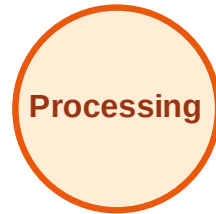
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing



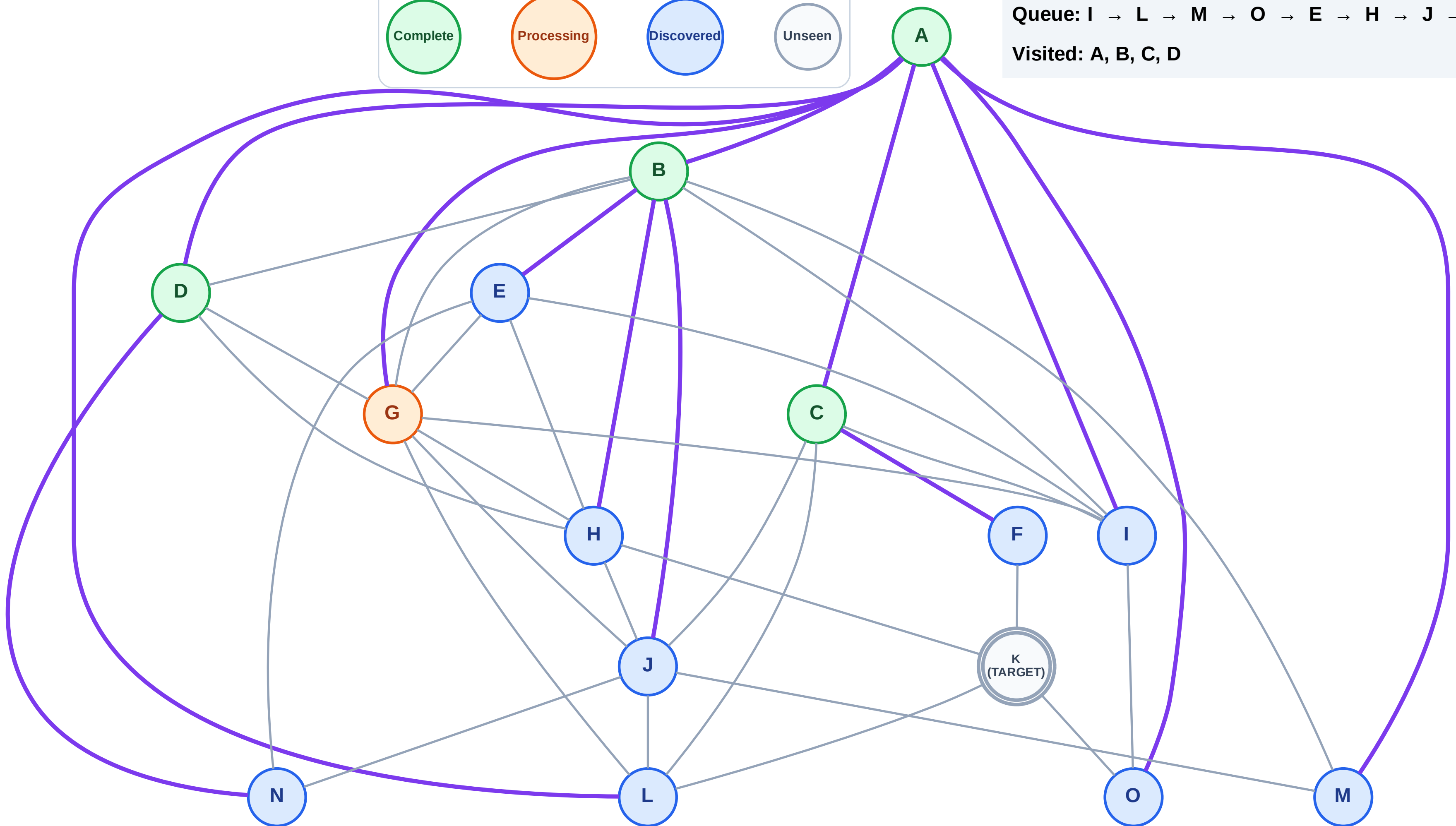
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

Visited: A, B, C, D



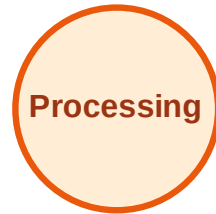
BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



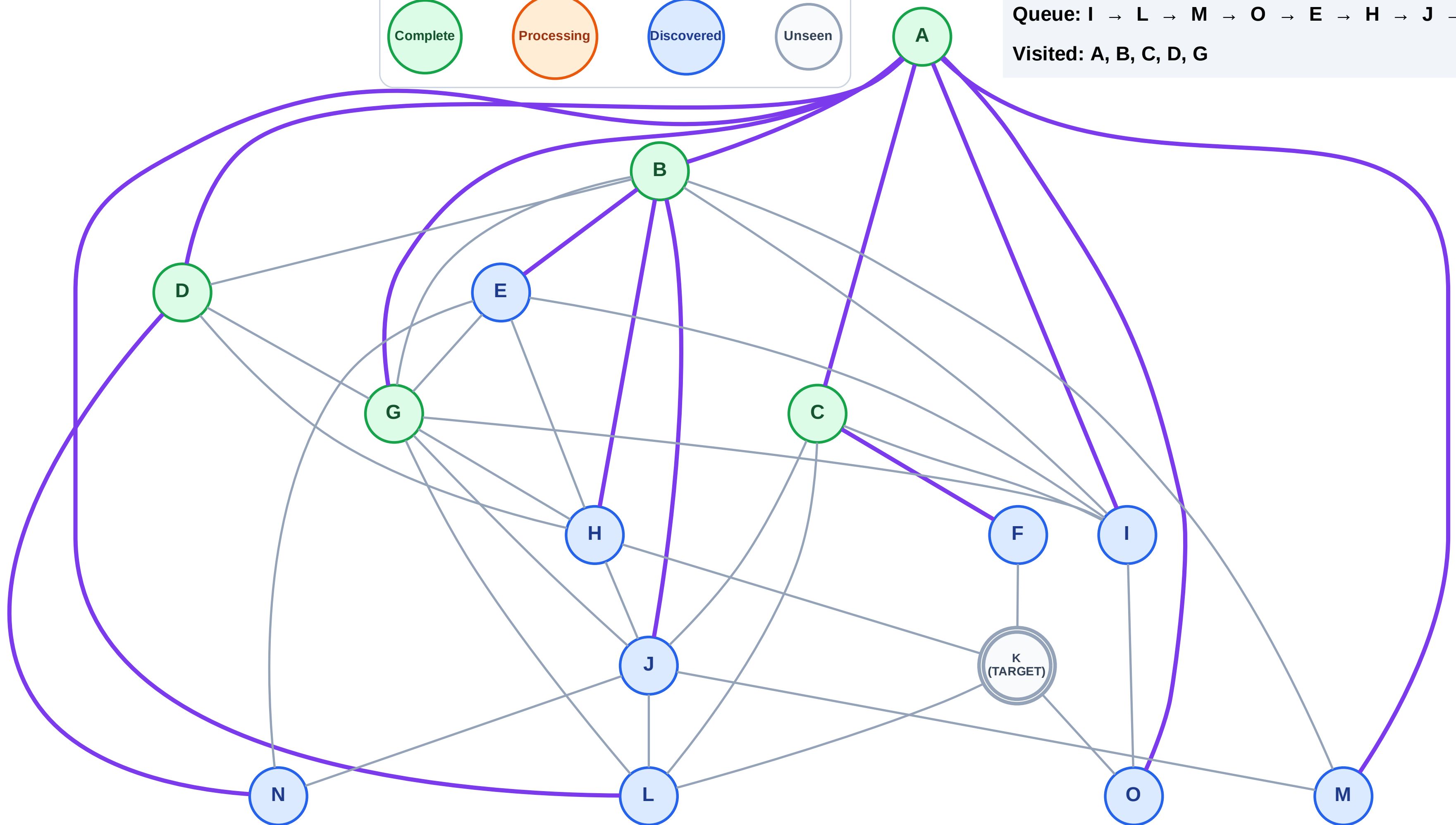
Discovered



Unseen

Queue: I → L → M → O → E → H → J → F → N

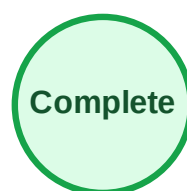
Visited: A, B, C, D, G



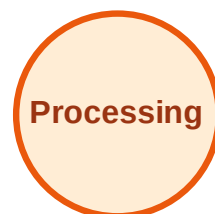
BFS Traversal

Step 12: Process node I

Legend



Complete



Processing



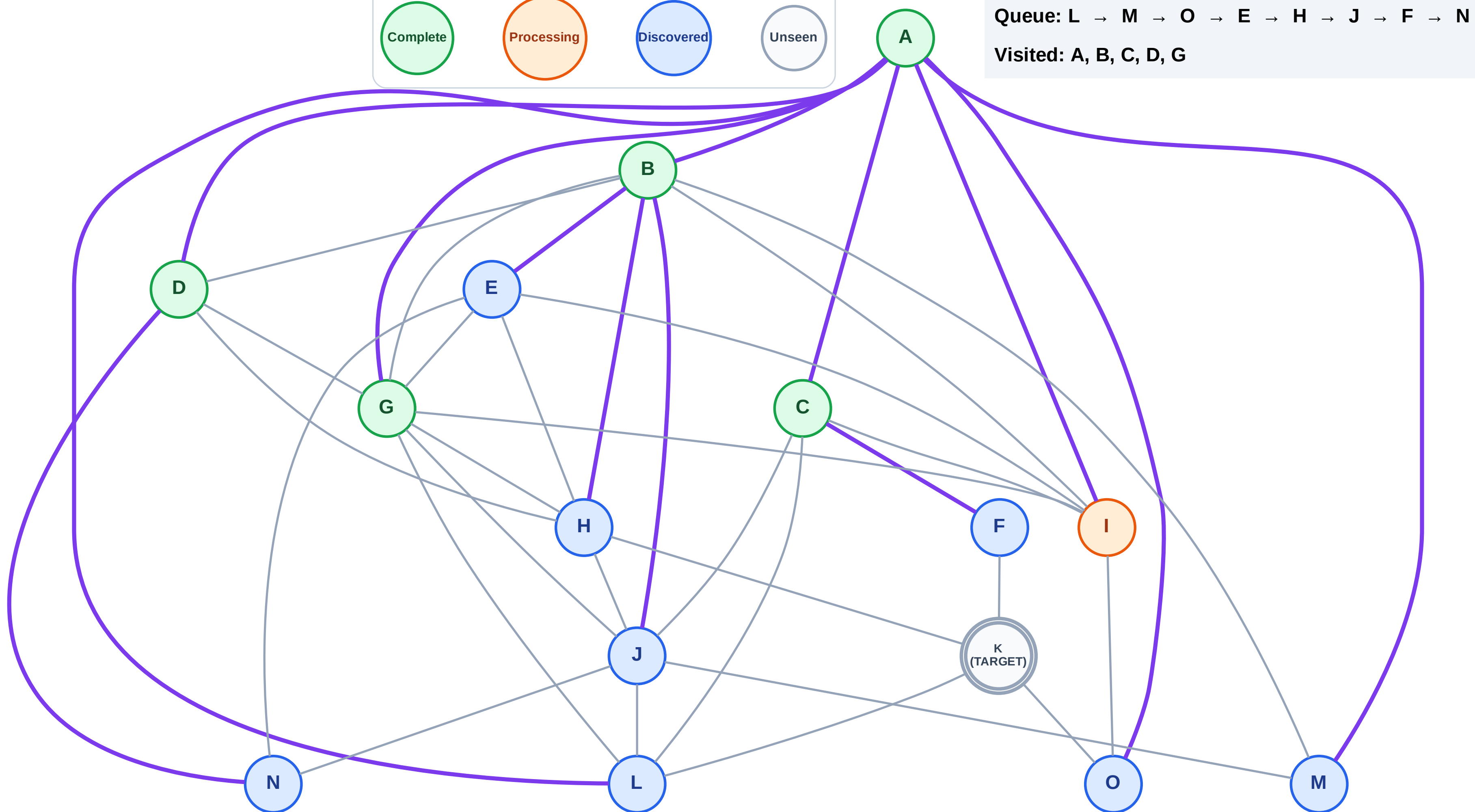
Discovered



Unseen

Queue: L → M → O → E → H → J → F → N

Visited: A, B, C, D, G



BFS Traversal

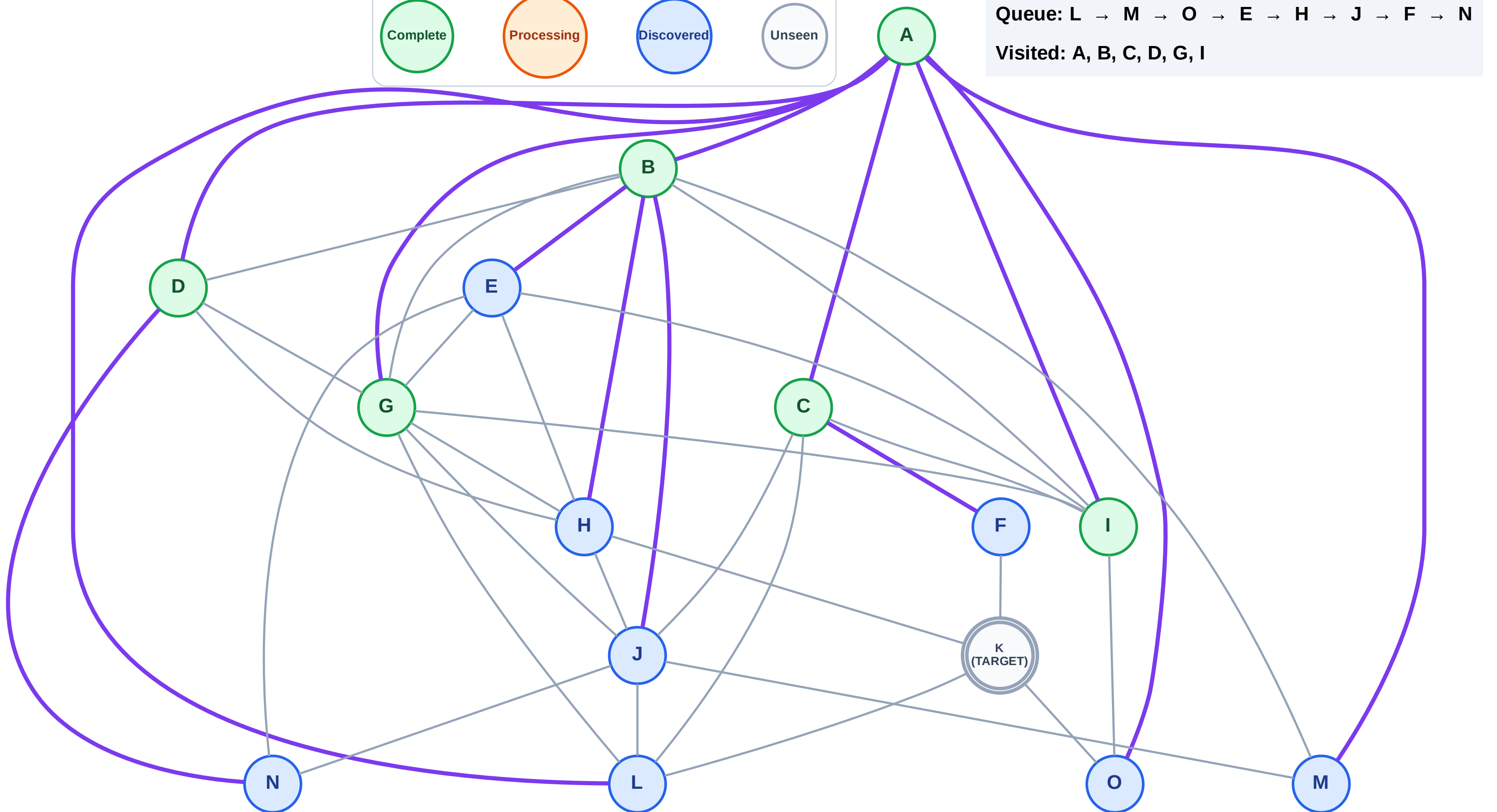
Step 13: No new neighbors from I

Legend



Queue: L → M → O → E → H → J → F → N

Visited: A, B, C, D, G, I



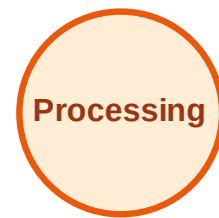
BFS Traversal

Step 14: Process node L

Legend



Complete



Processing



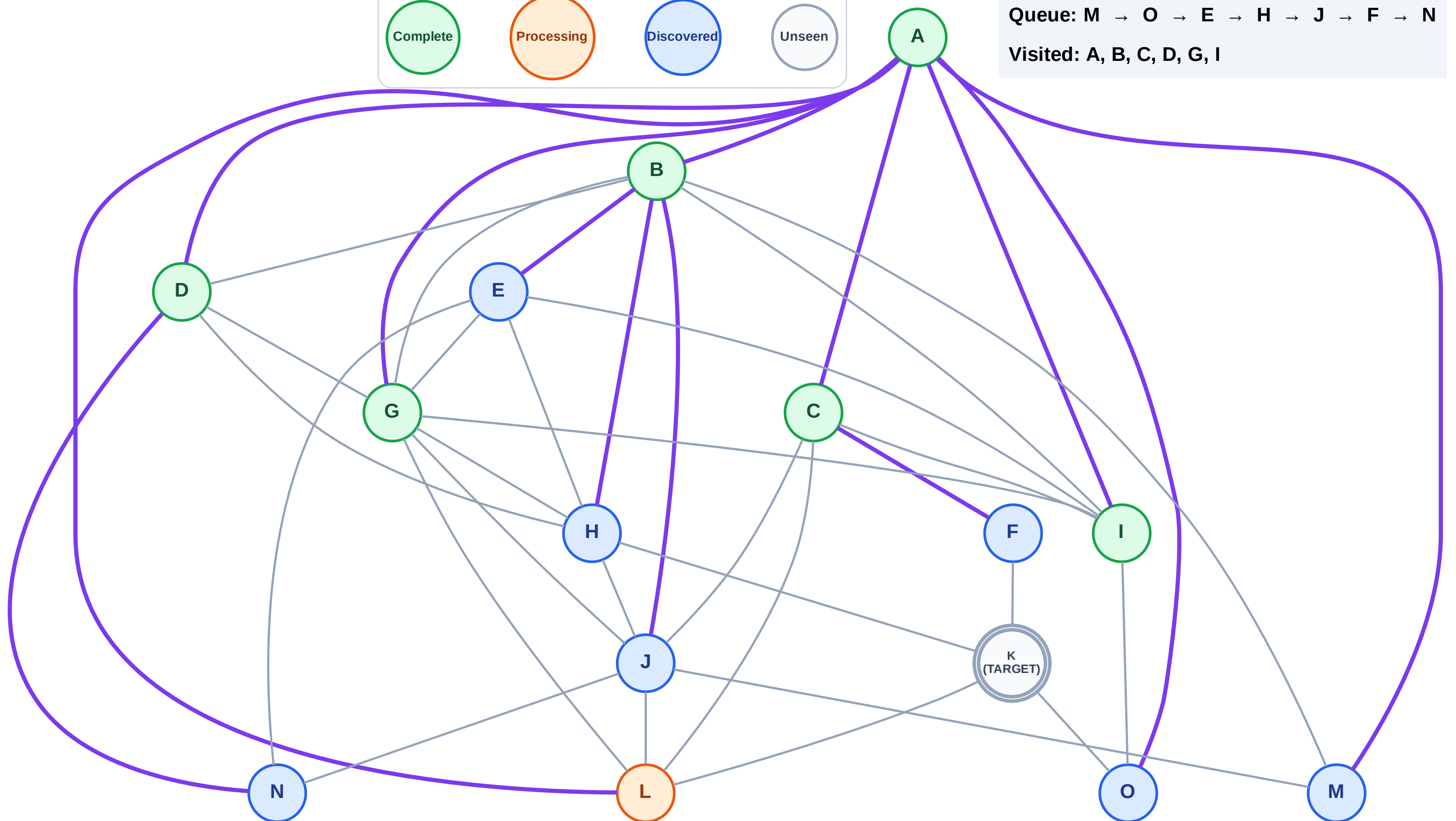
Discovered



Unseen

Queue: M → O → E → H → J → F → N

Visited: A, B, C, D, G, I



BFS Traversal

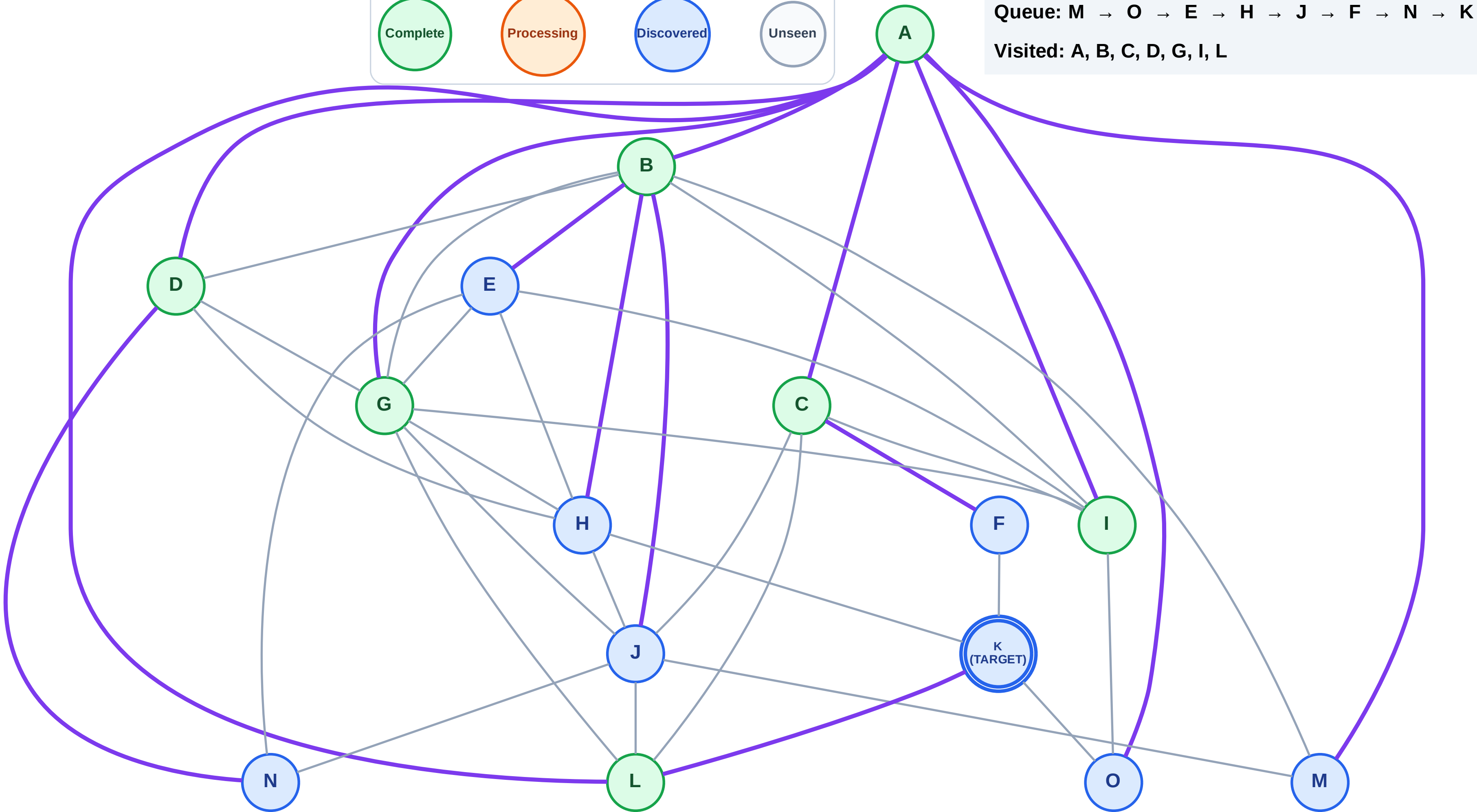
Step 15: Discover K from L

Legend



Queue: M → O → E → H → J → F → N → K

Visited: A, B, C, D, G, I, L



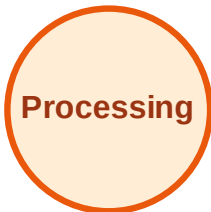
BFS Traversal

Step 16: Process node M

Legend



Complete



Processing



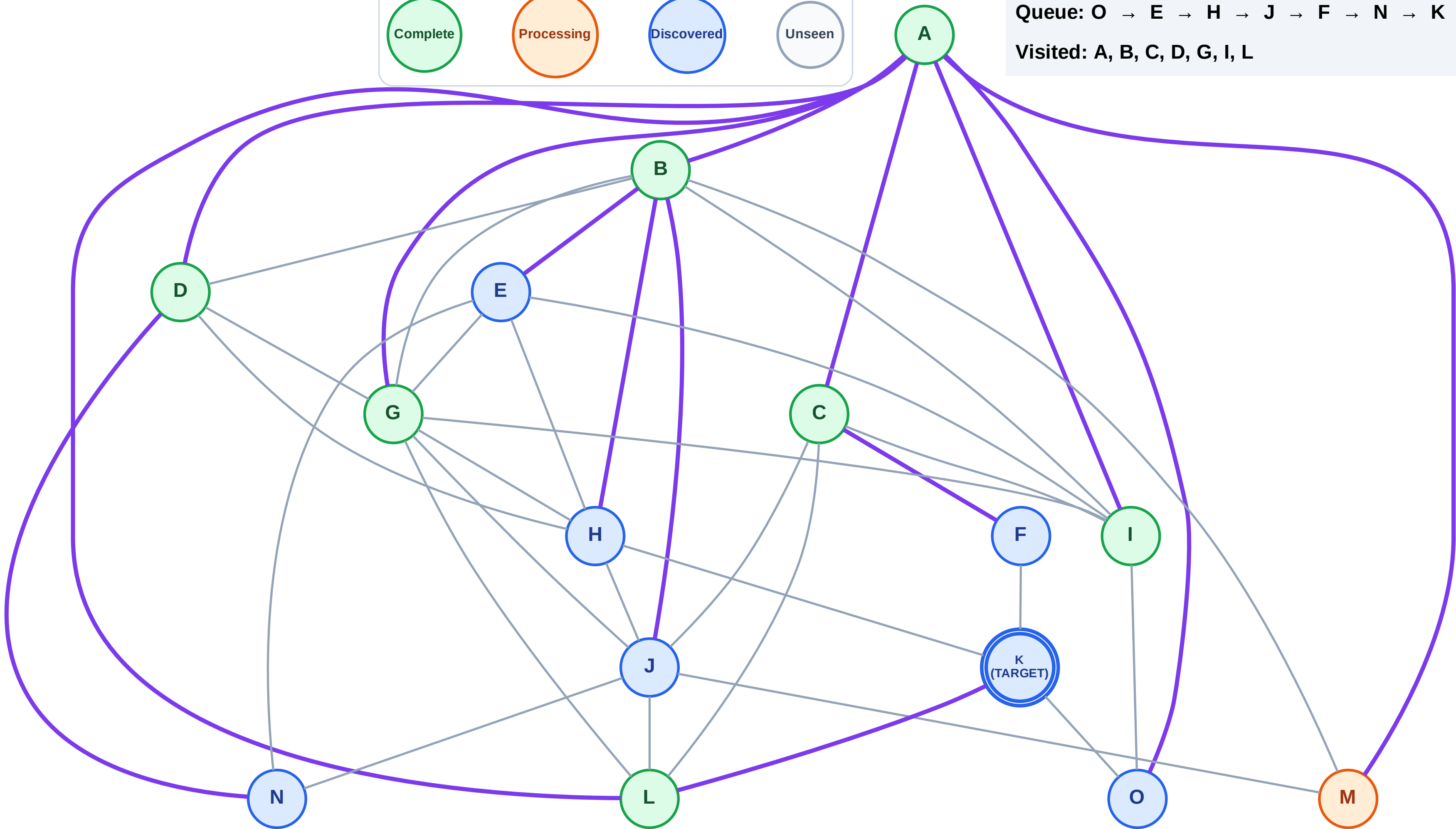
Discovered



Unseen

Queue: O → E → H → J → F → N → K

Visited: A, B, C, D, G, I, L



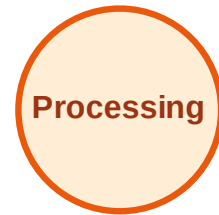
BFS Traversal

Step 17: No new neighbors from M

Legend



Complete



Processing



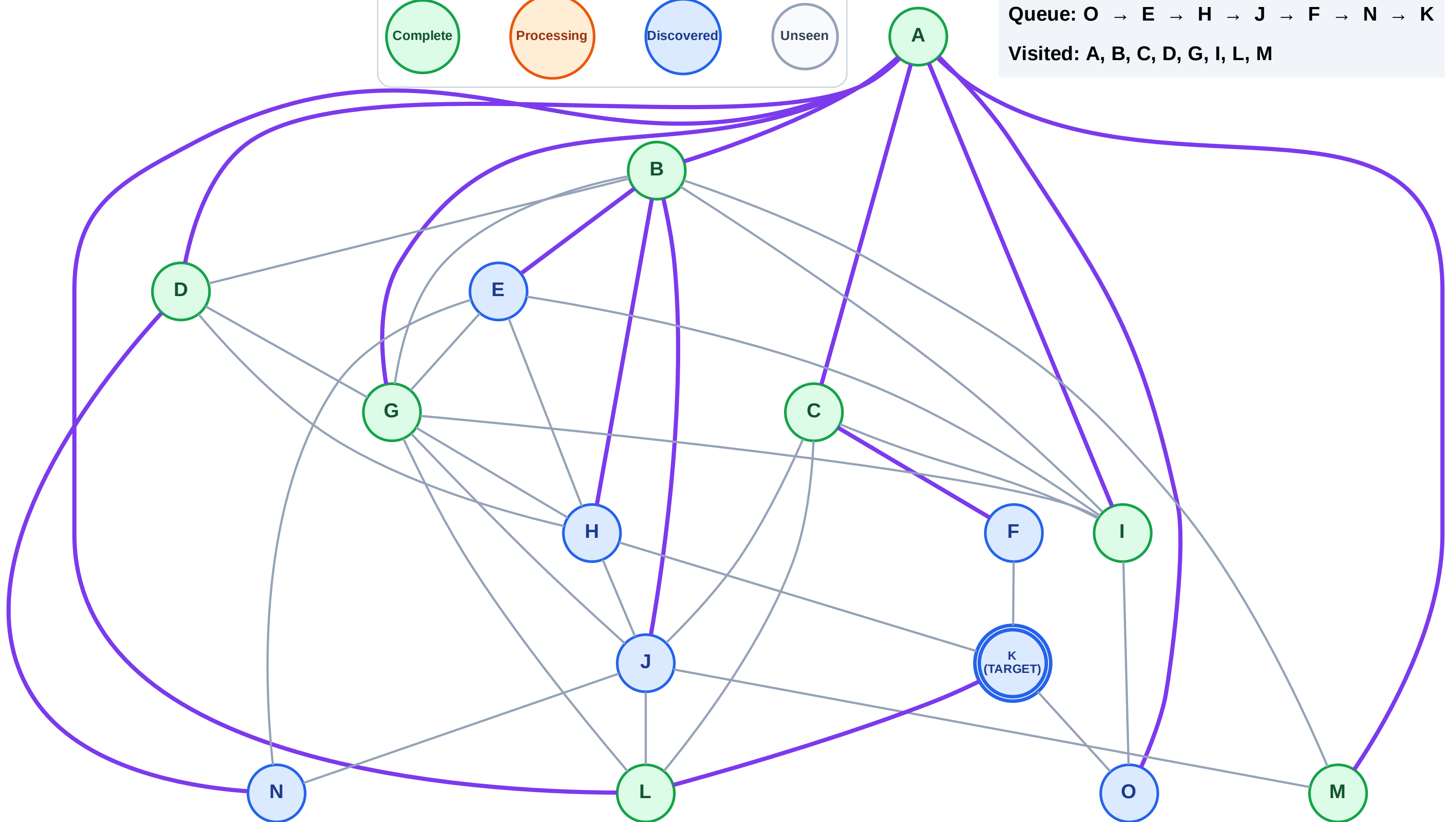
Discovered



Unseen

Queue: O → E → H → J → F → N → K

Visited: A, B, C, D, G, I, L, M



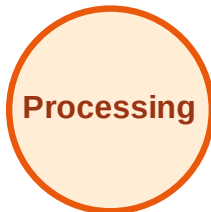
BFS Traversal

Step 18: Process node O

Legend



Complete



Processing



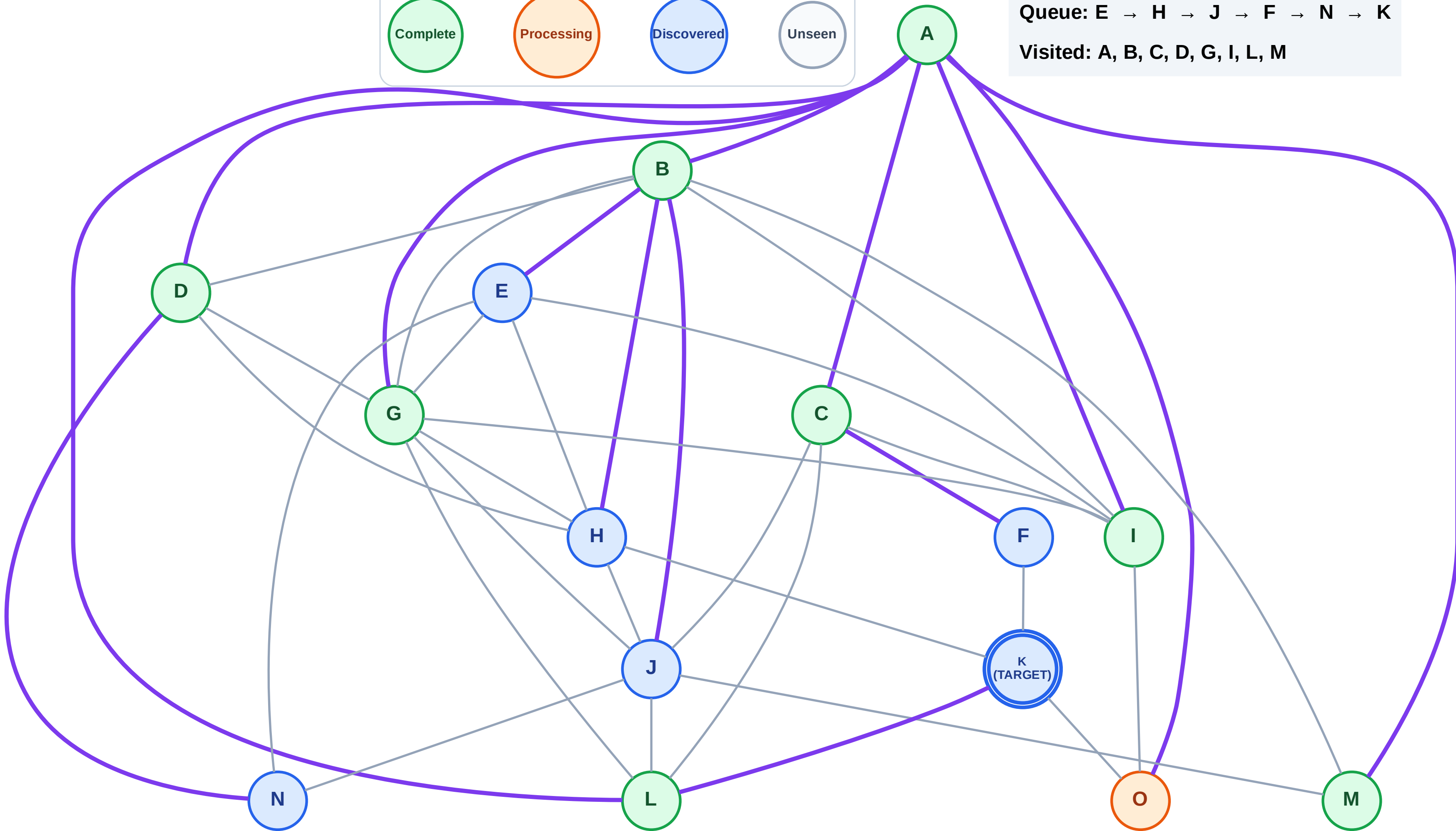
Discovered



Unseen

Queue: E → H → J → F → N → K

Visited: A, B, C, D, G, I, L, M



BFS Traversal

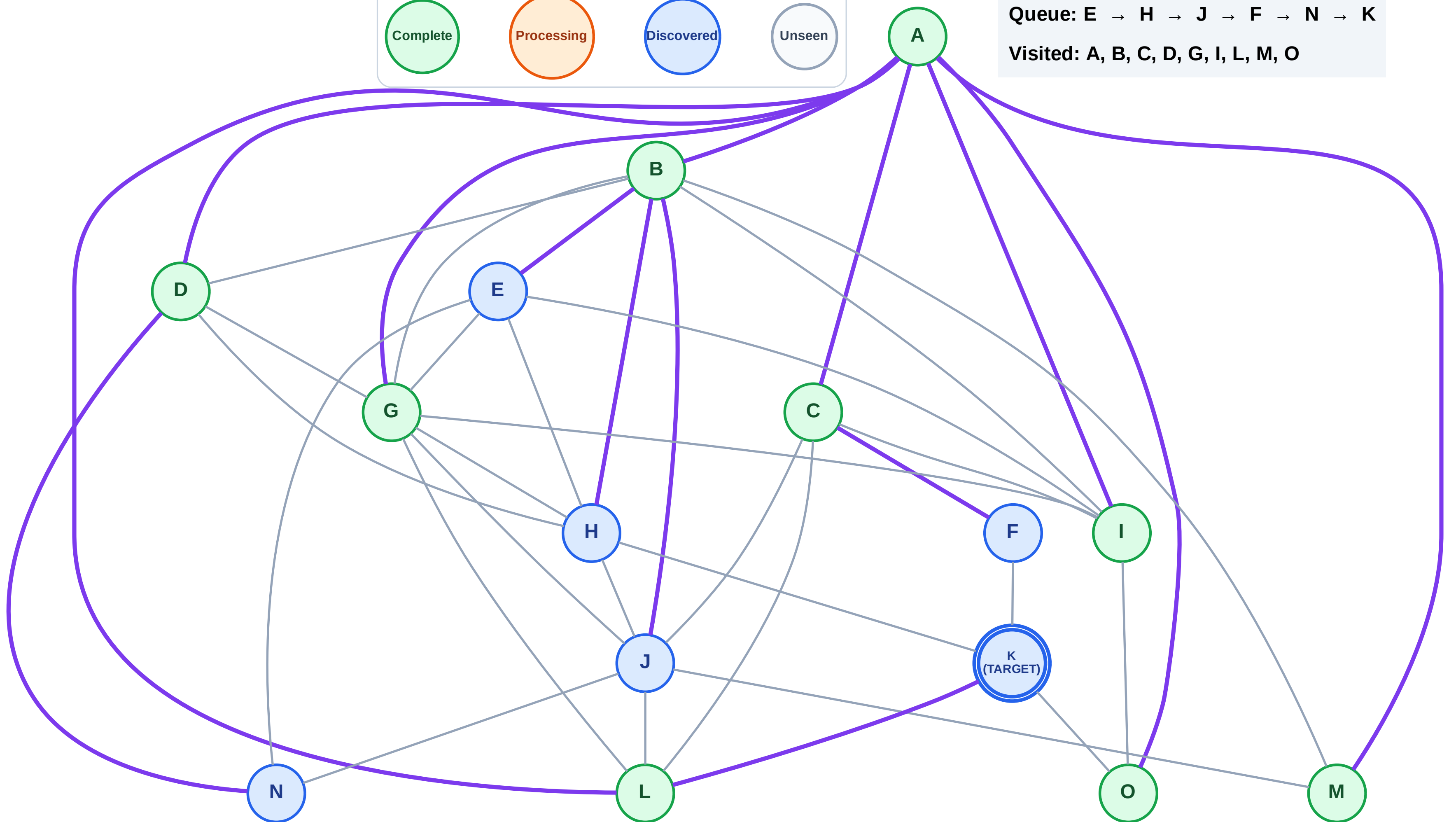
Step 19: No new neighbors from O

Legend



Queue: E → H → J → F → N → K

Visited: A, B, C, D, G, I, L, M, O



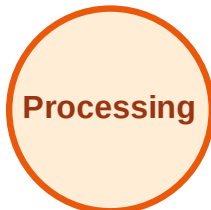
BFS Traversal

Step 20: Process node E

Legend



Complete



Processing



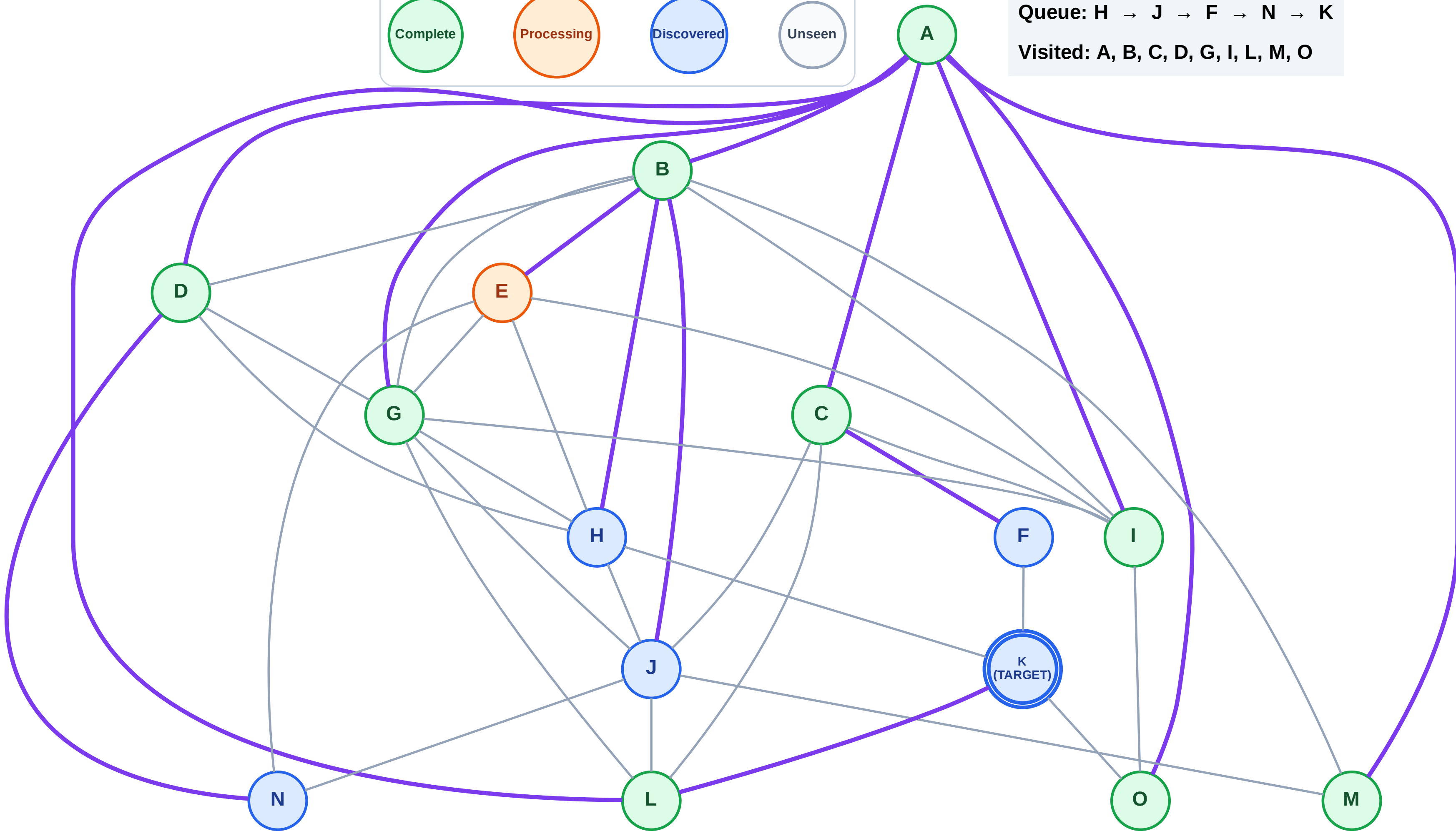
Discovered



Unseen

Queue: H → J → F → N → K

Visited: A, B, C, D, G, I, L, M, O



BFS Traversal

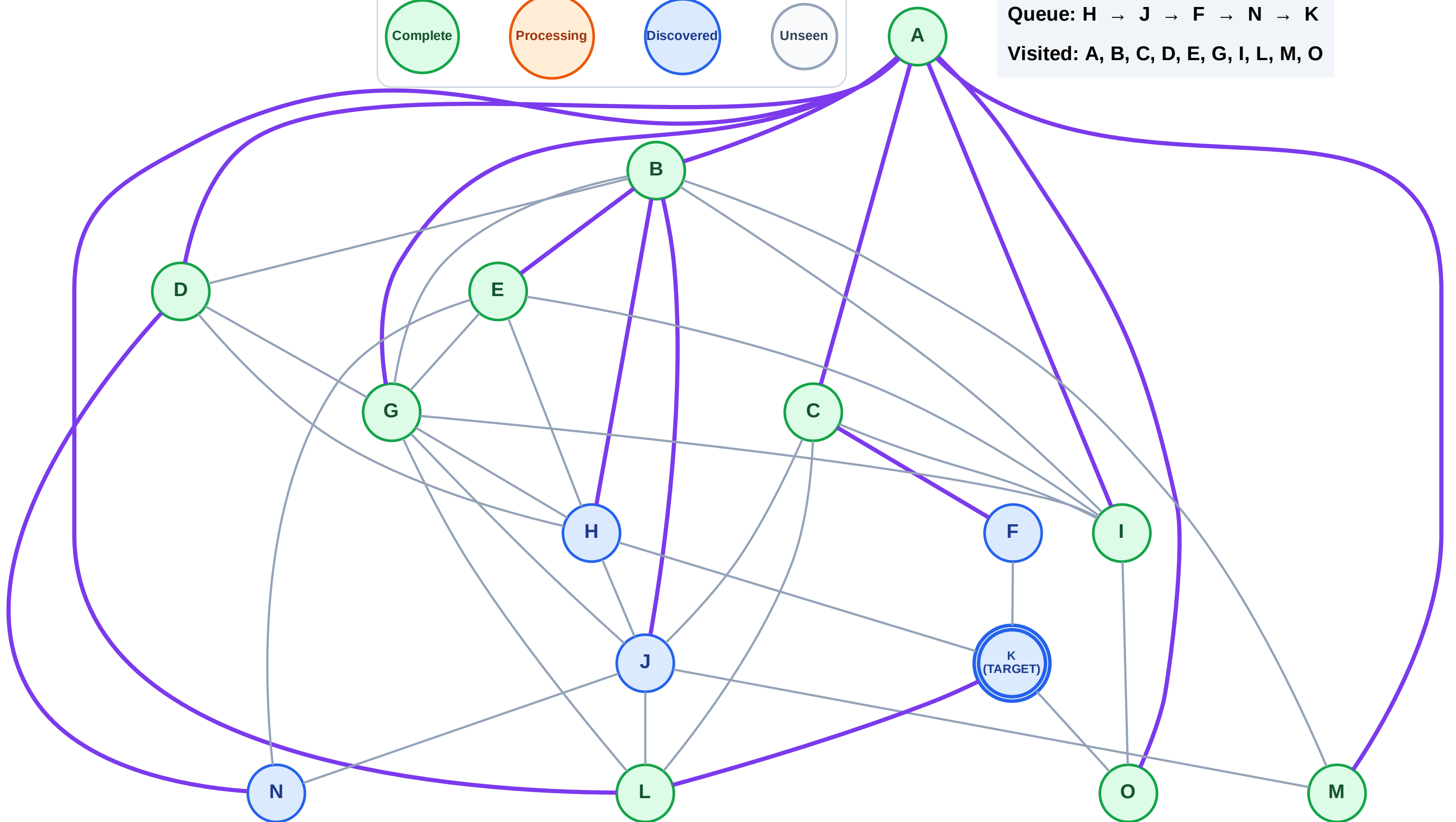
Step 21: No new neighbors from E

Legend



Queue: H → J → F → N → K

Visited: A, B, C, D, E, G, I, L, M, O



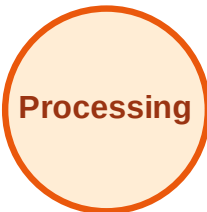
BFS Traversal

Step 22: Process node H

Legend



Complete



Processing



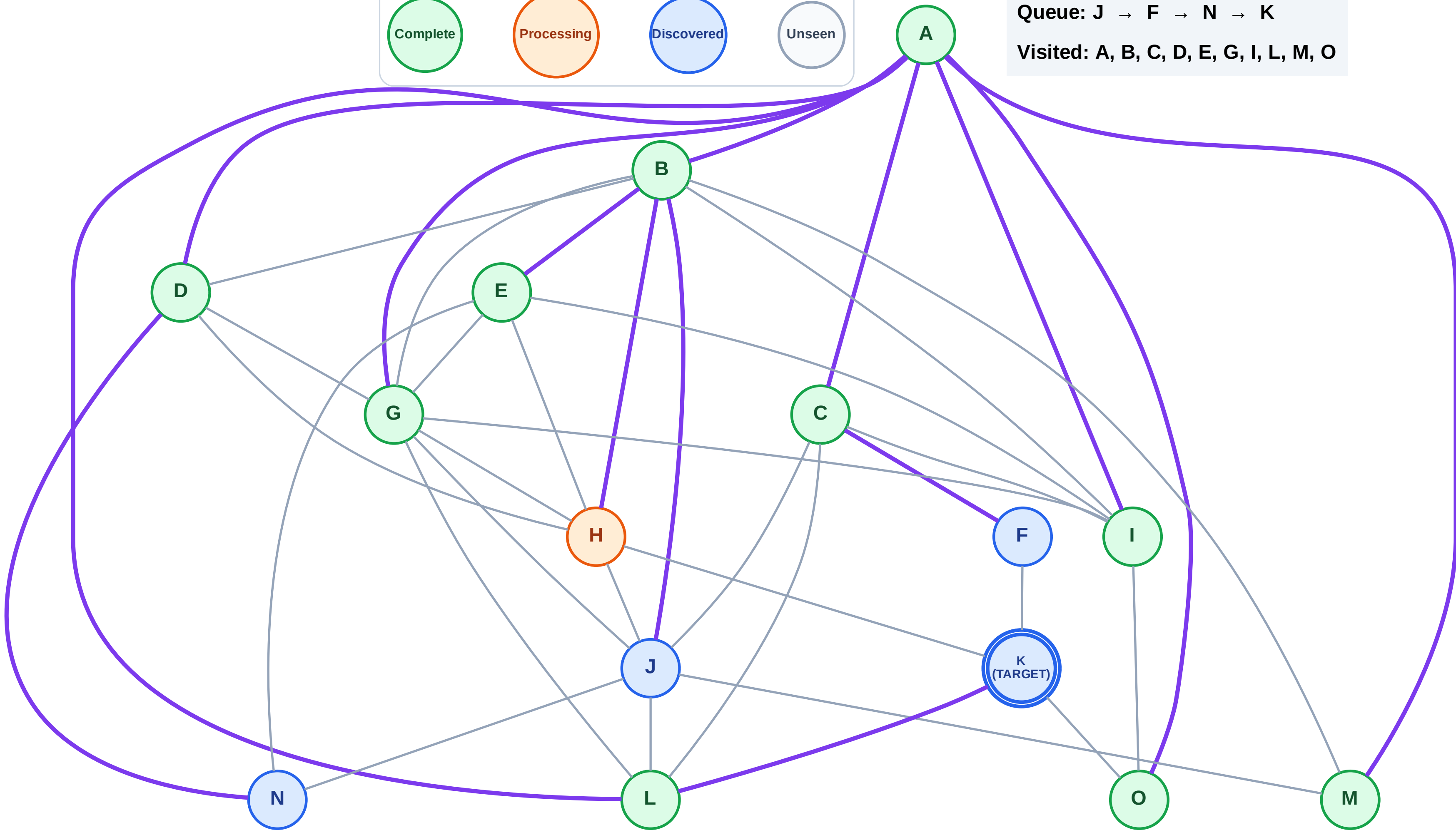
Discovered



Unseen

Queue: J → F → N → K

Visited: A, B, C, D, E, G, I, L, M, O



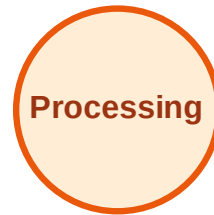
BFS Traversal

Step 23: No new neighbors from H

Legend



Complete



Processing



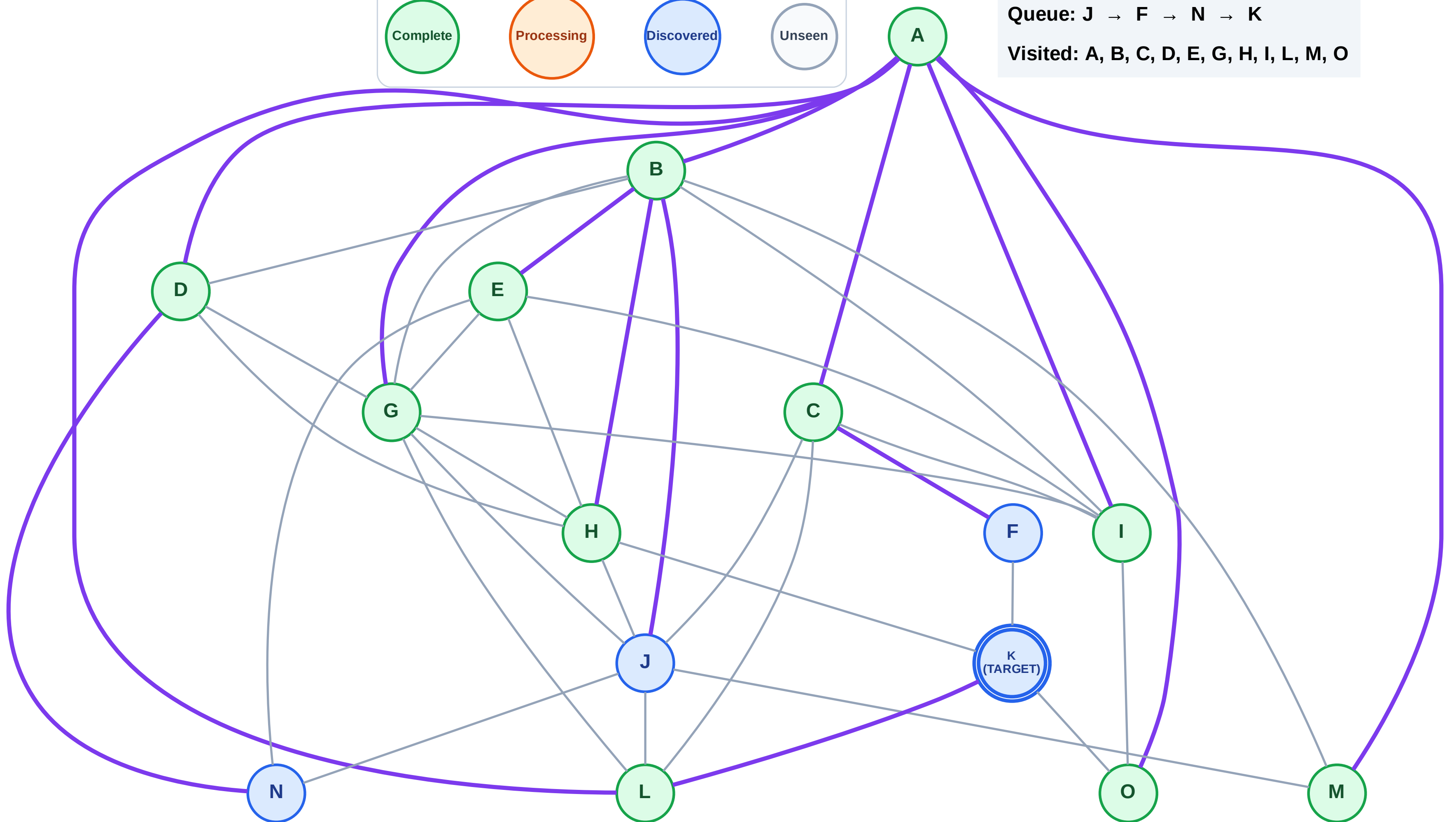
Discovered



Unseen

Queue: J → F → N → K

Visited: A, B, C, D, E, G, H, I, L, M, O



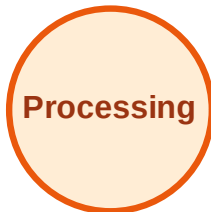
BFS Traversal

Step 24: Process node J

Legend



Complete



Processing



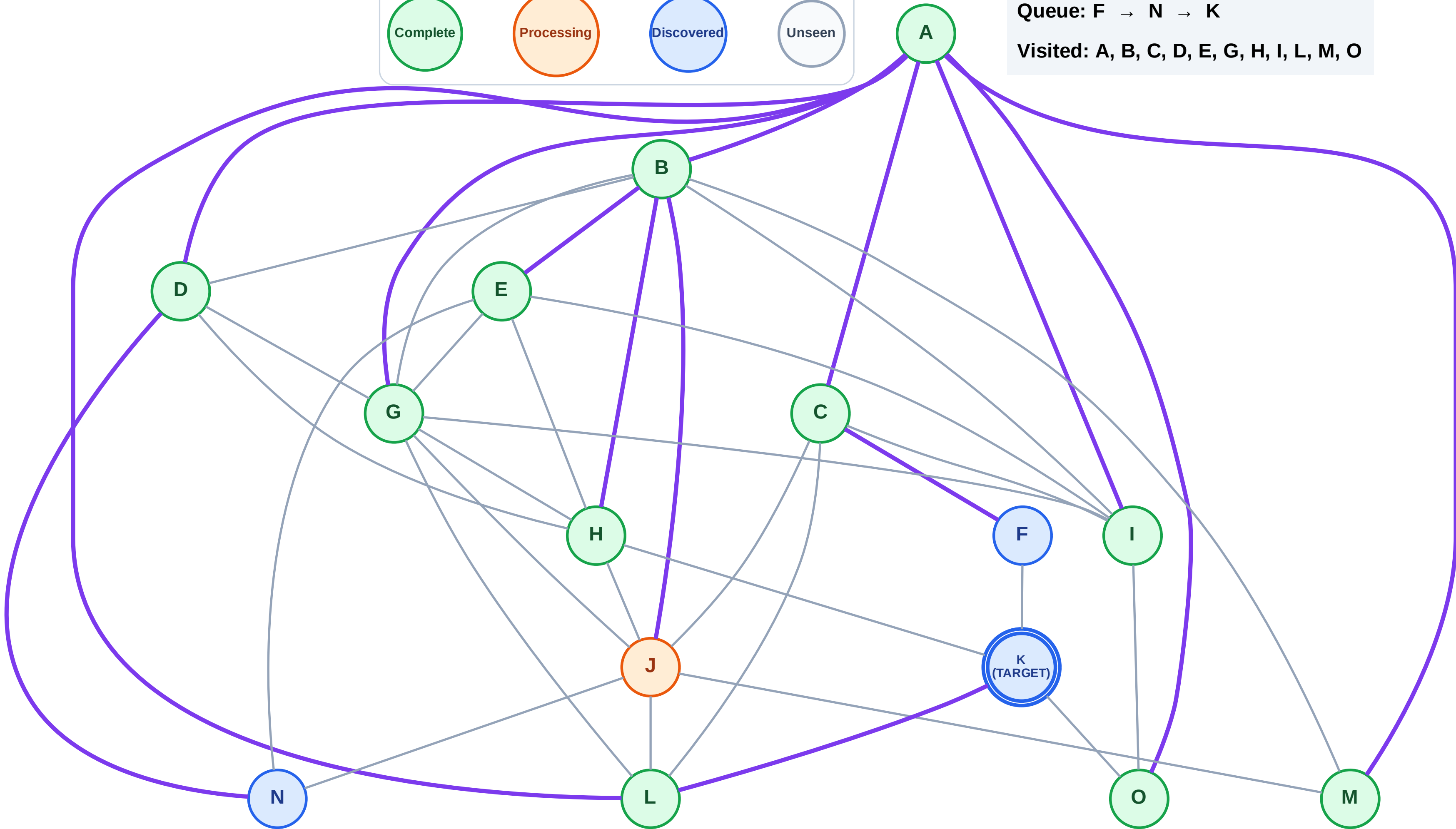
Discovered



Unseen

Queue: F → N → K

Visited: A, B, C, D, E, G, H, I, L, M, O



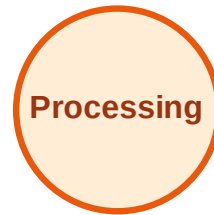
BFS Traversal

Step 25: No new neighbors from J

Legend



Complete



Processing



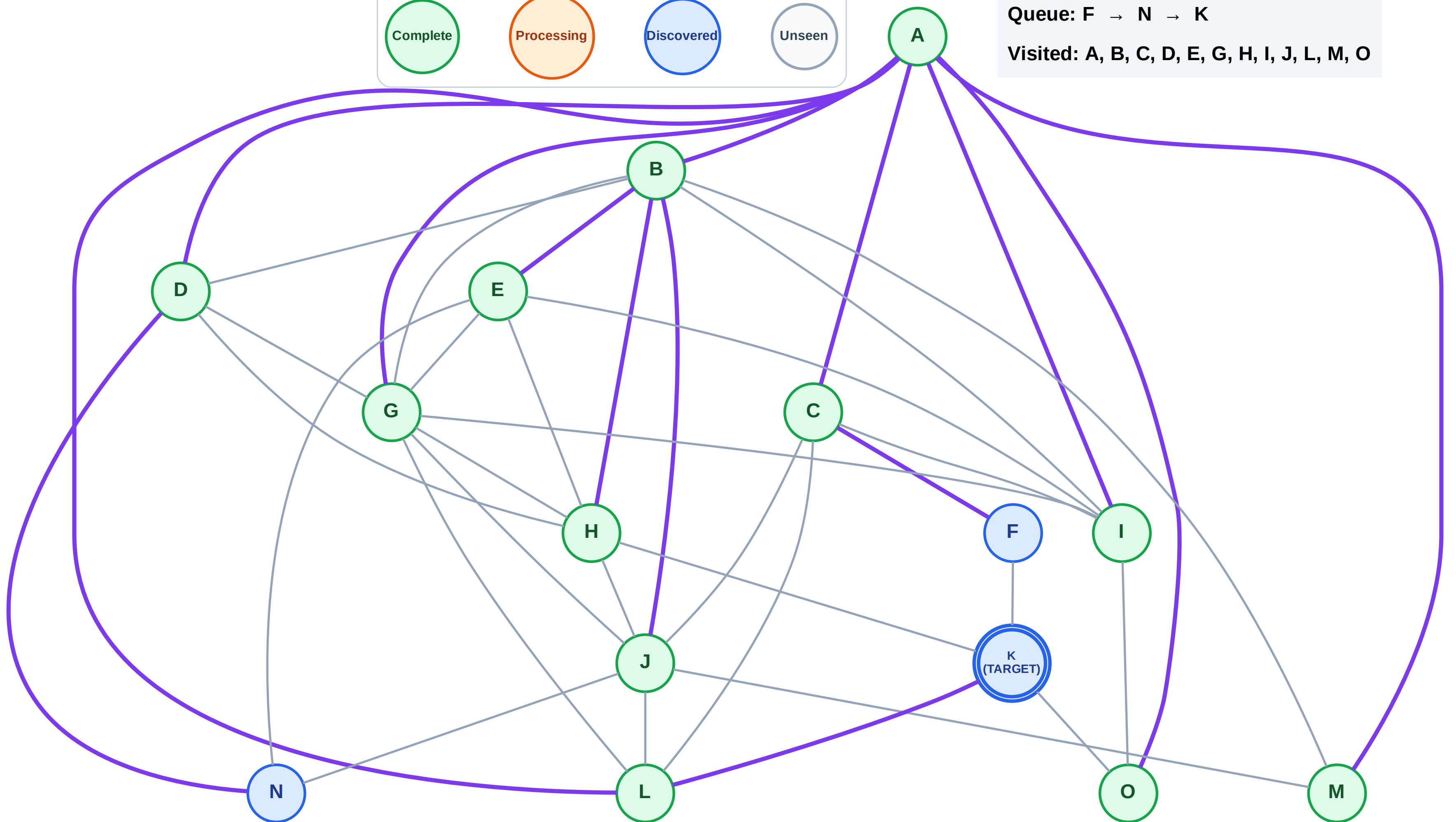
Discovered



Unseen

Queue: F → N → K

Visited: A, B, C, D, E, G, H, I, J, L, M, O



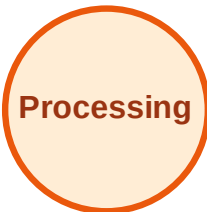
BFS Traversal

Step 26: Process node F

Legend



Complete



Processing



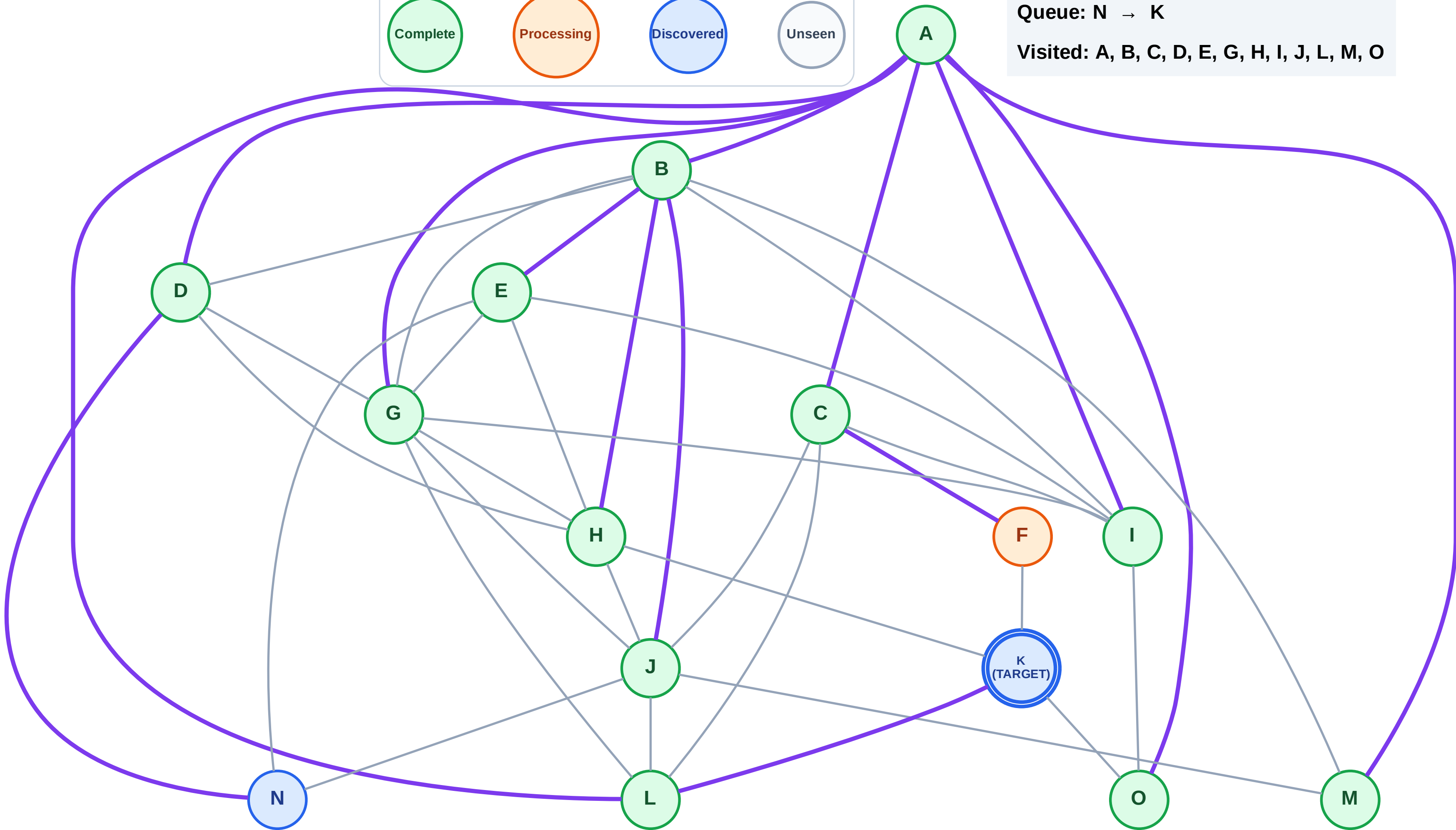
Discovered



Unseen

Queue: N → K

Visited: A, B, C, D, E, G, H, I, J, L, M, O



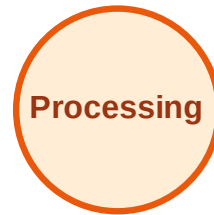
BFS Traversal

Step 27: No new neighbors from F

Legend



Complete



Processing



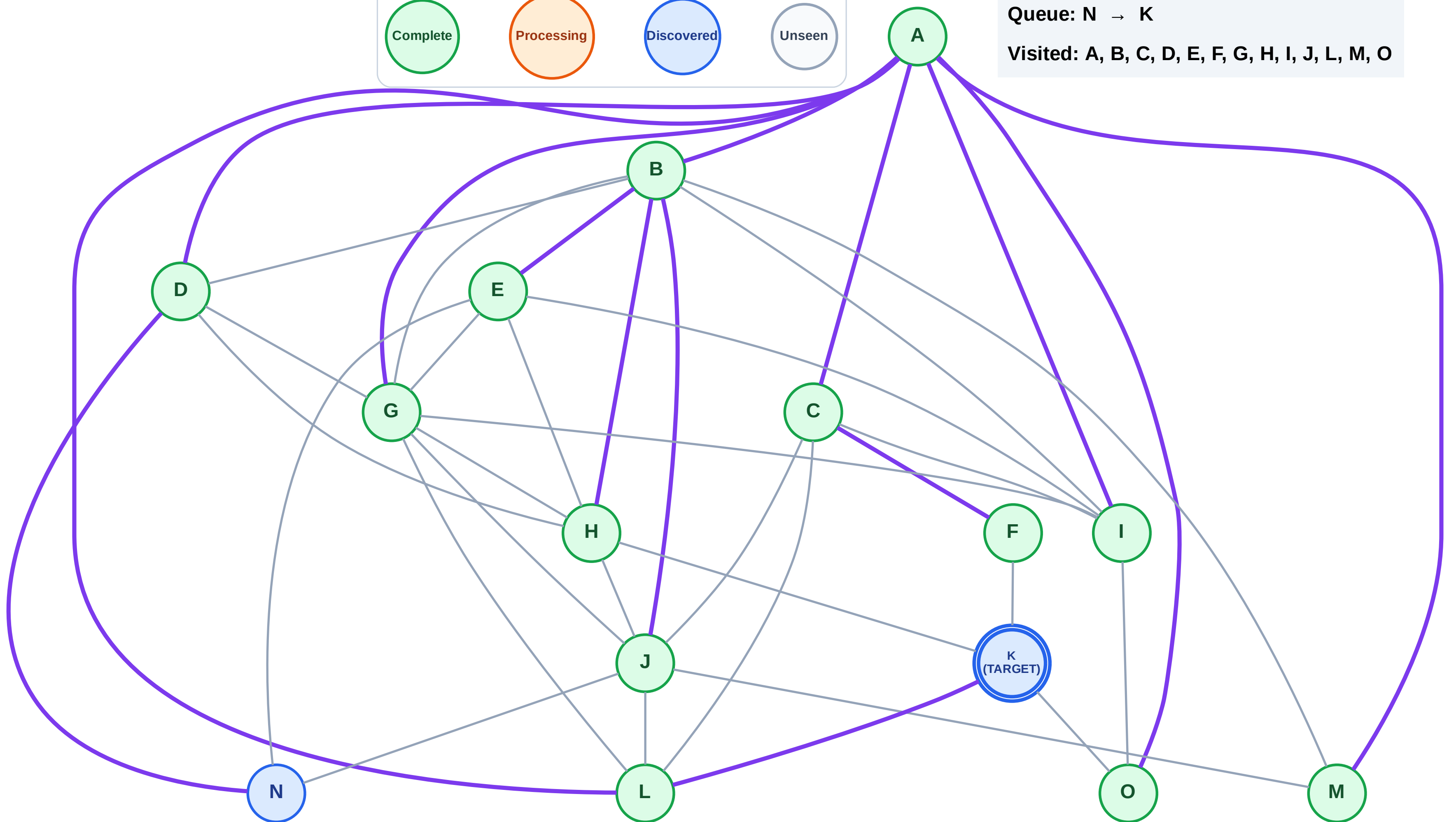
Discovered



Unseen

Queue: N → K

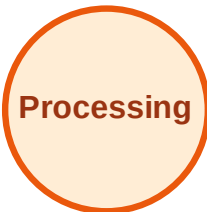
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

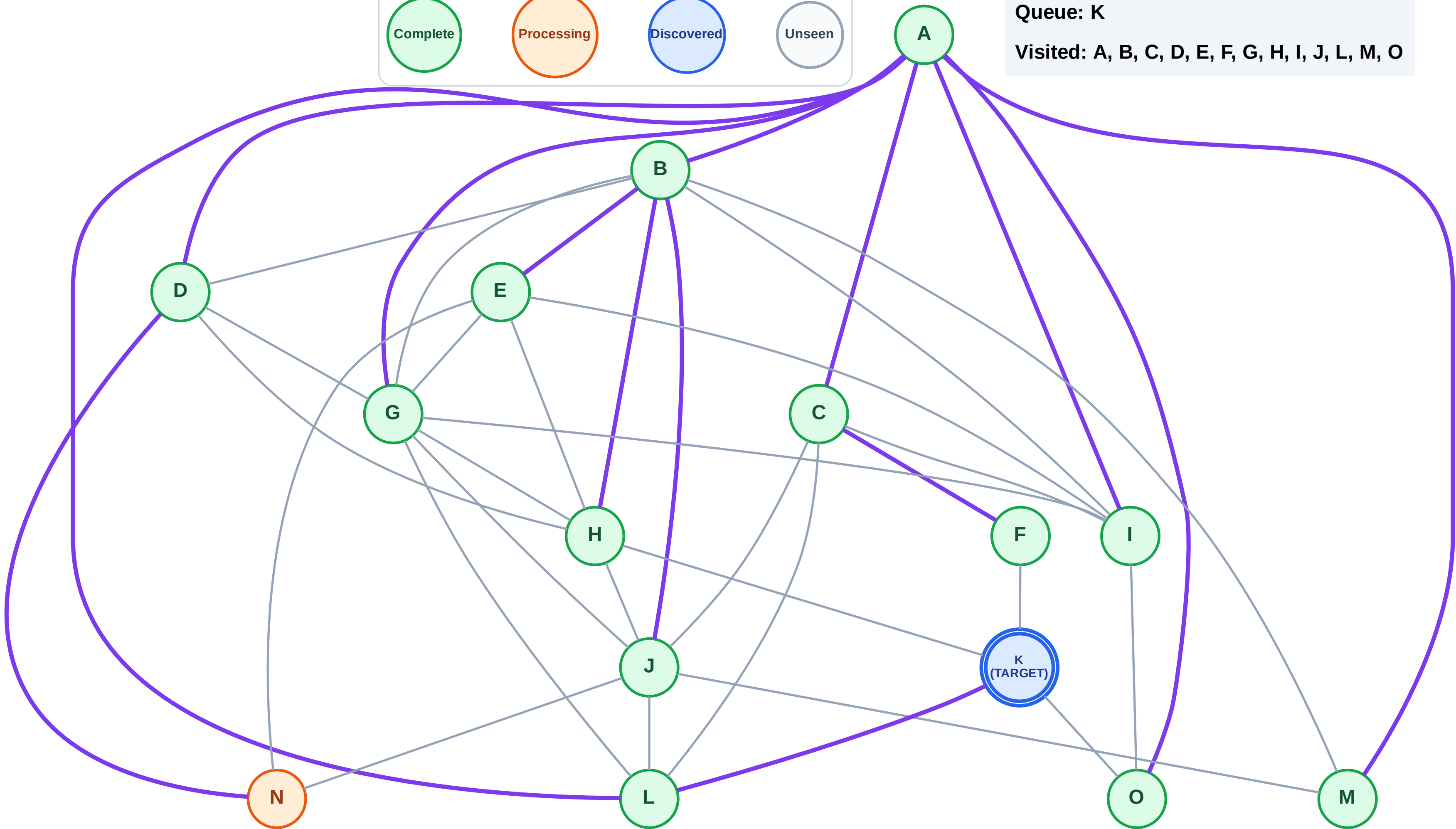
Step 28: Process node N

Legend



Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



Step 29: No new neighbors from N

Step 29: No new neighbors from N

Legend



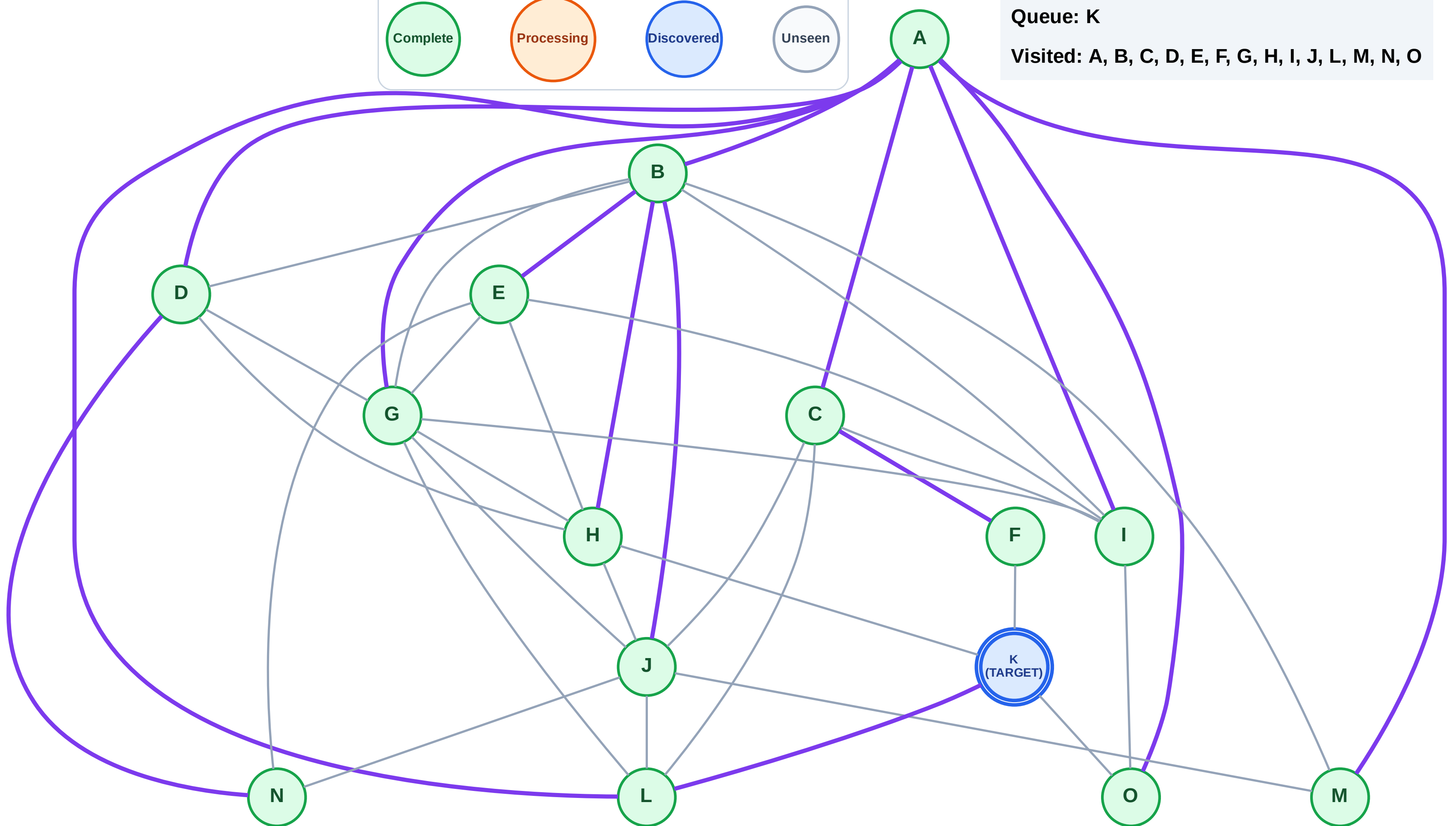
Complete Processing Discovered Unseen

Processing

Unseen

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O

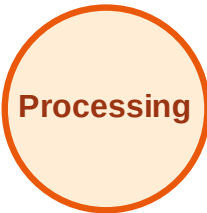
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

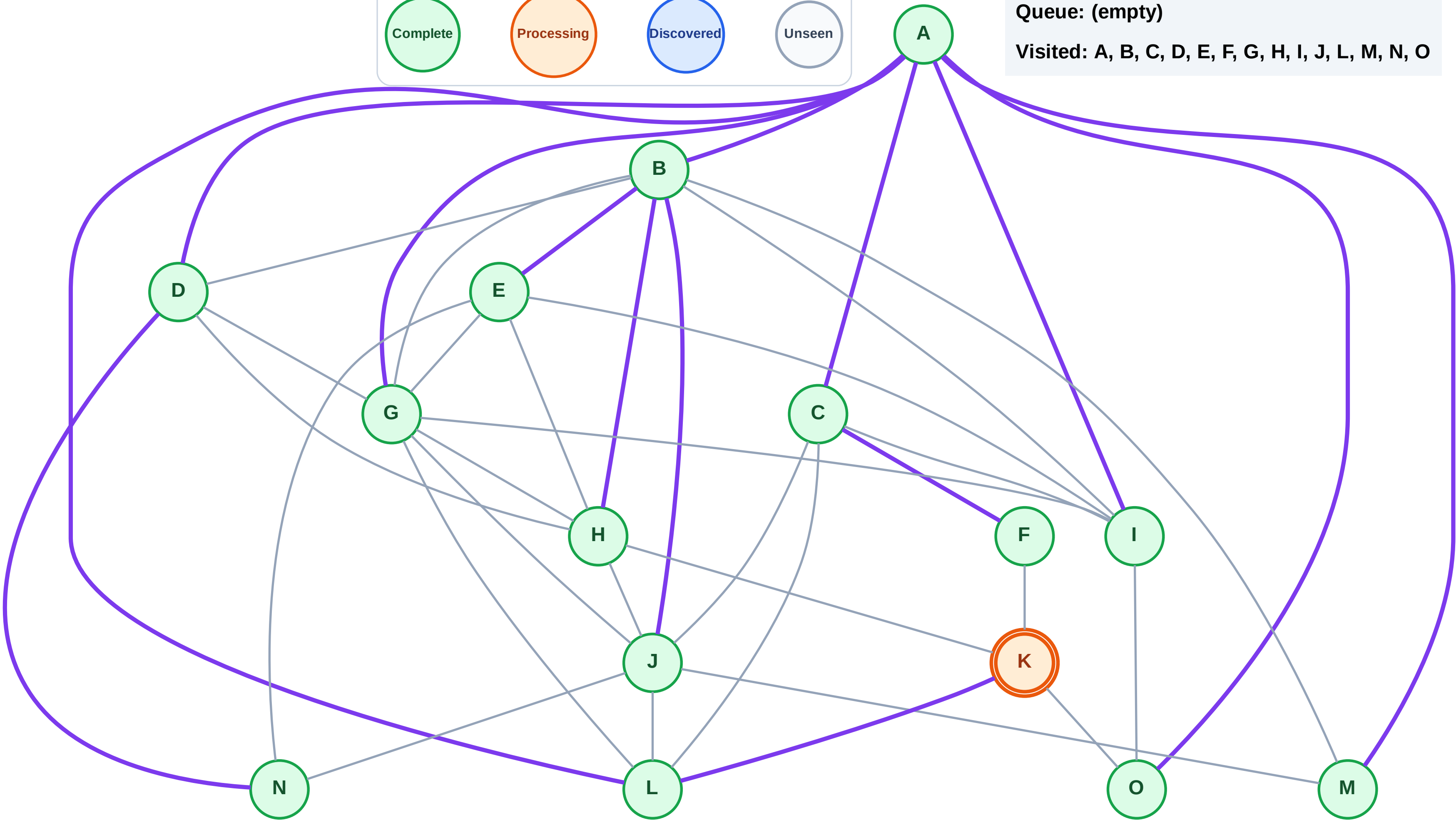
Step 30: Process node K

Legend



Queue: (empty)

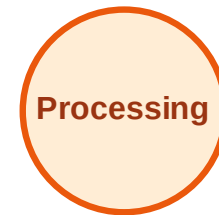
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

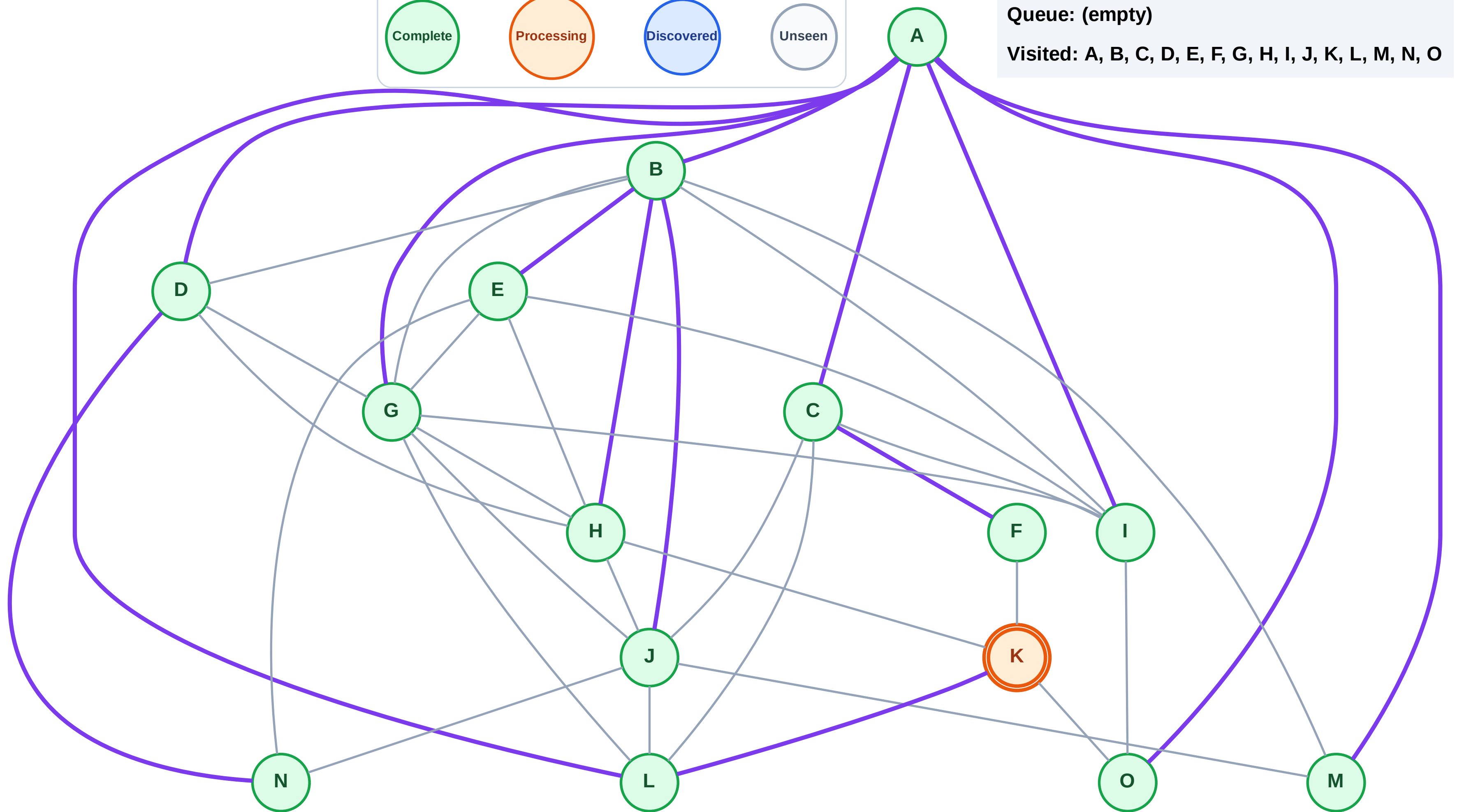
Step 31: TARGET NODE K REACHED! Path (2 hops): A → L → K

Legend



Queue: (empty)

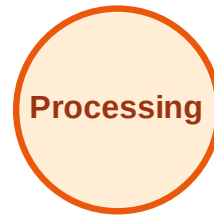
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

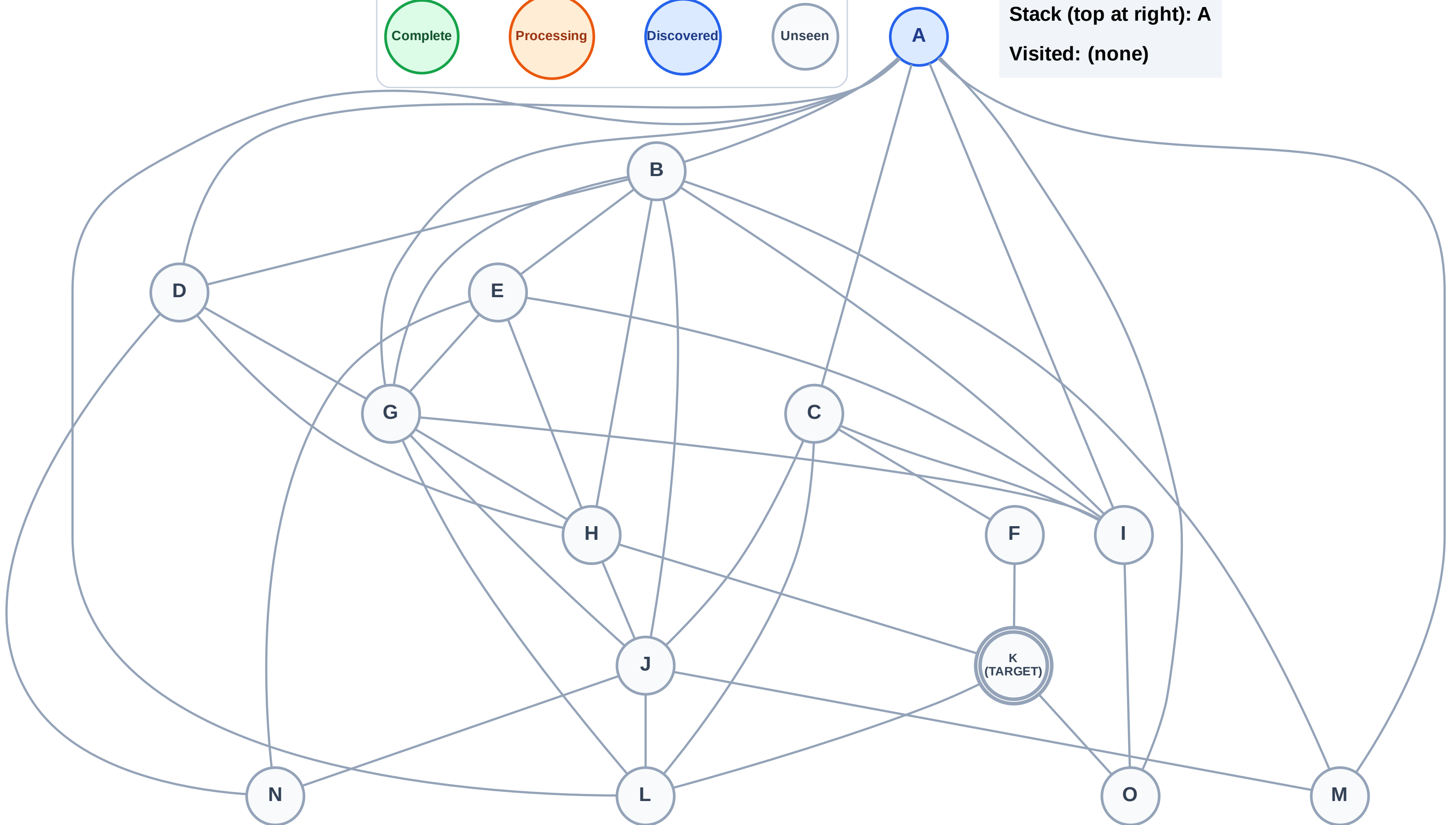
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

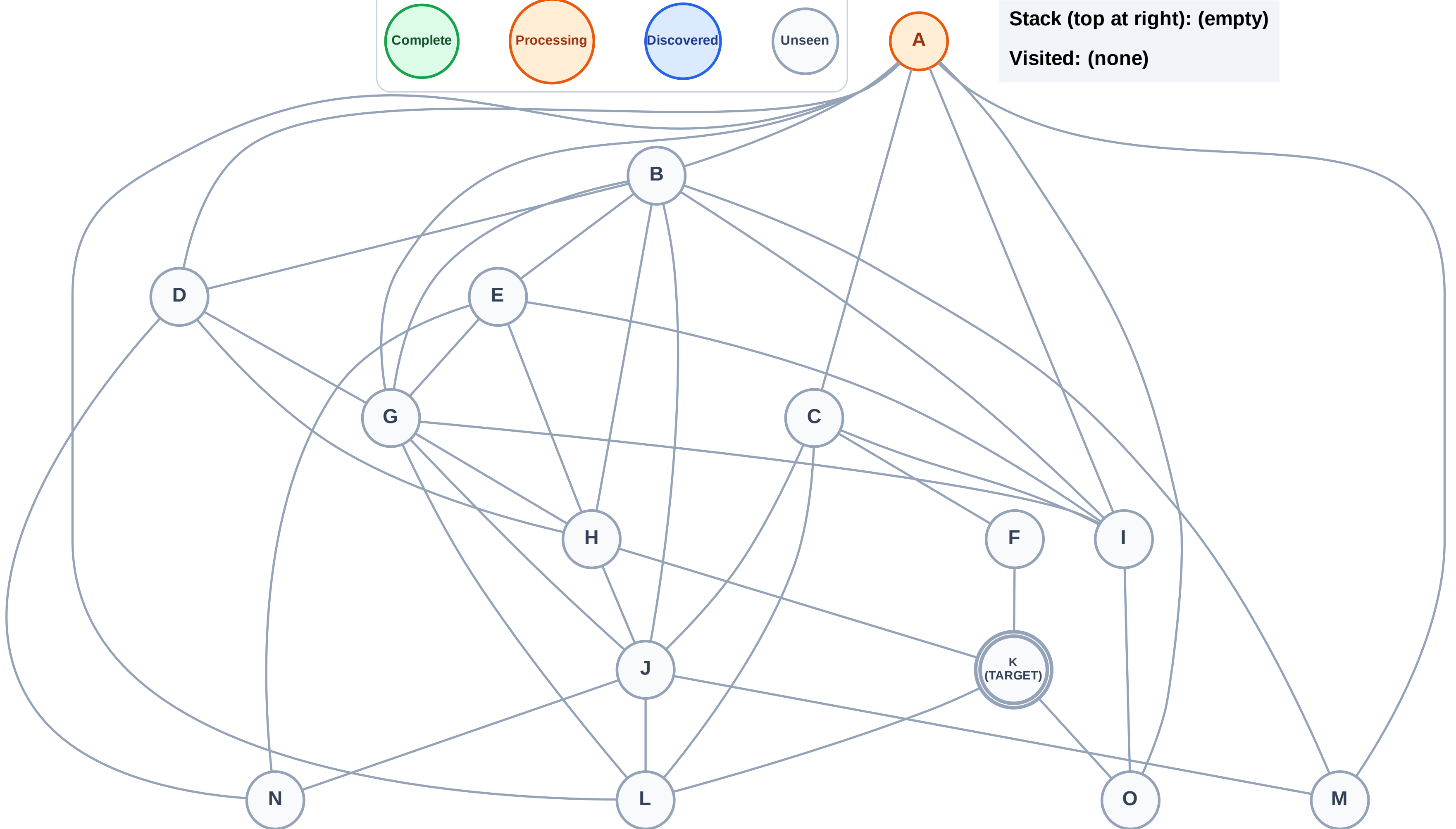
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

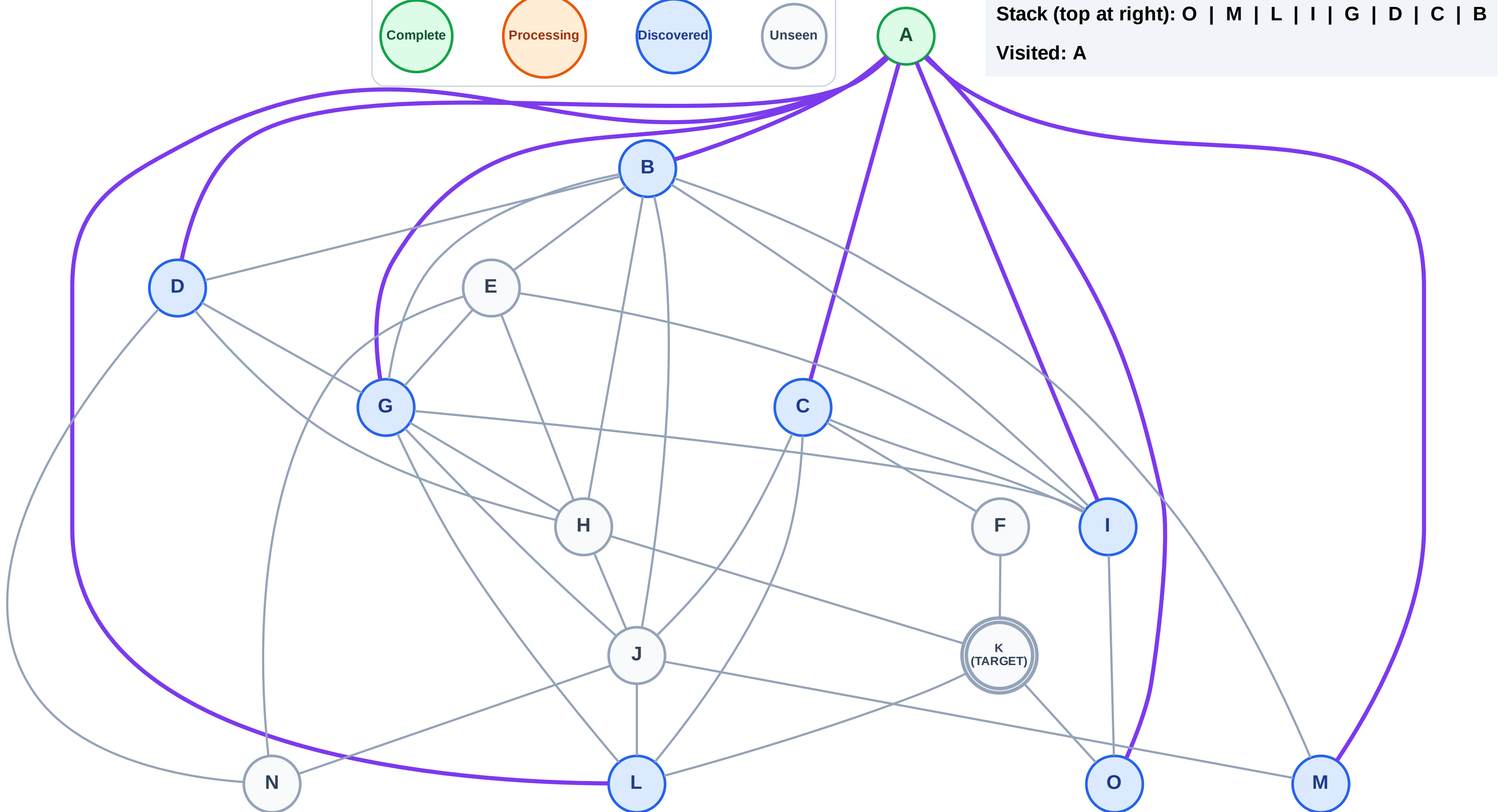
Step 3: Discover O, M, L, I, G, D, C, B from A

Legend



Stack (top at right): O | M | L | I | G | D | C | B

Visited: A



DFS Traversal

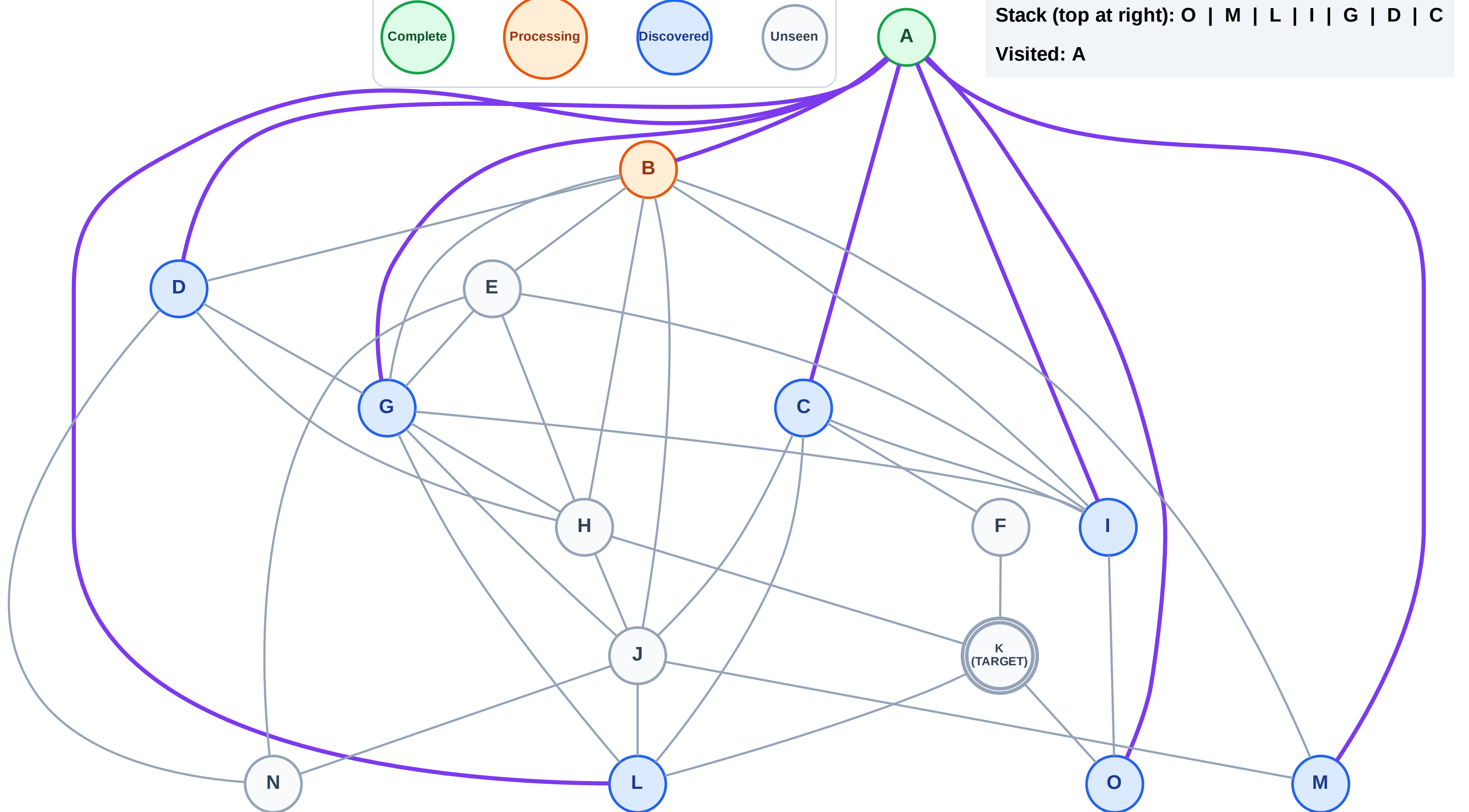
Step 4: Process node B

Legend



Stack (top at right): O | M | L | I | G | D | C

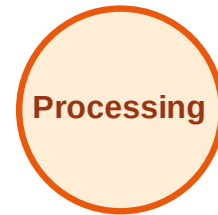
Visited: A



DFS Traversal

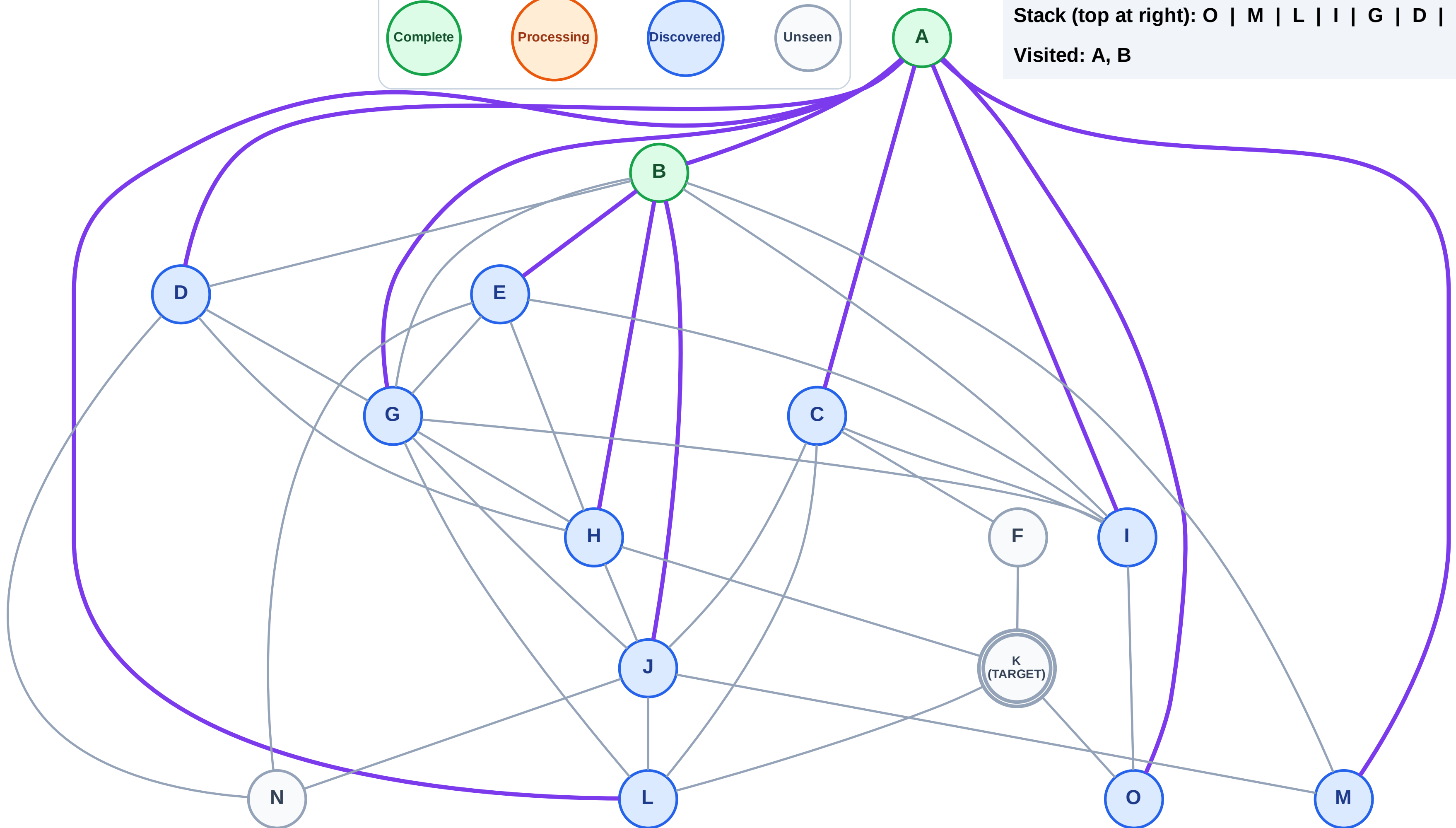
Step 5: Discover J, H, E from B

Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

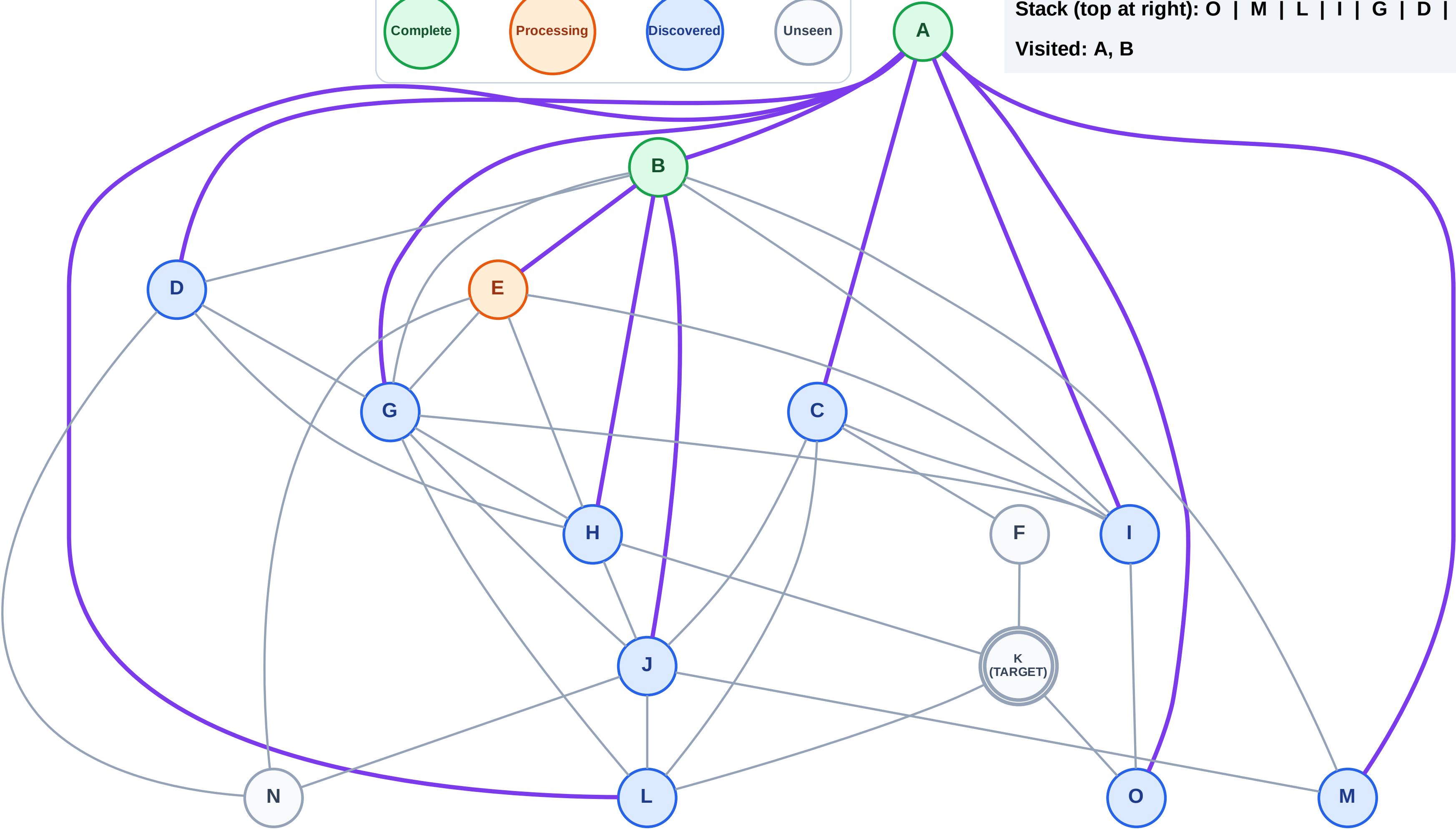
Processing

Discovered

Unseen

Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B



DFS Traversal

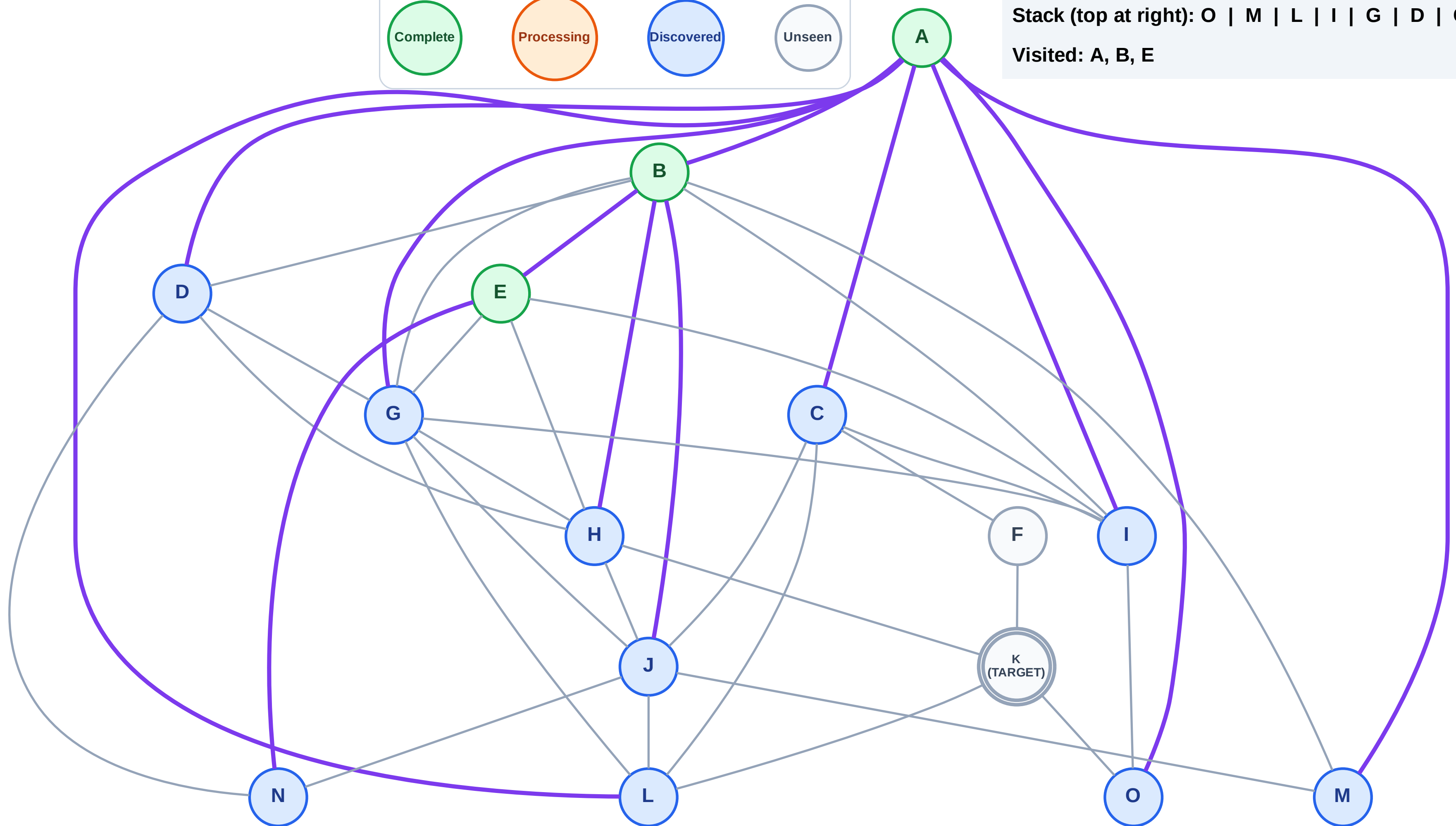
Step 7: Discover N from E

Legend



Stack (top at right): O | M | L | I | G | D | C | J | H | N

Visited: A, B, E



DFS Traversal

Step 8: Process node N

Legend

Complete

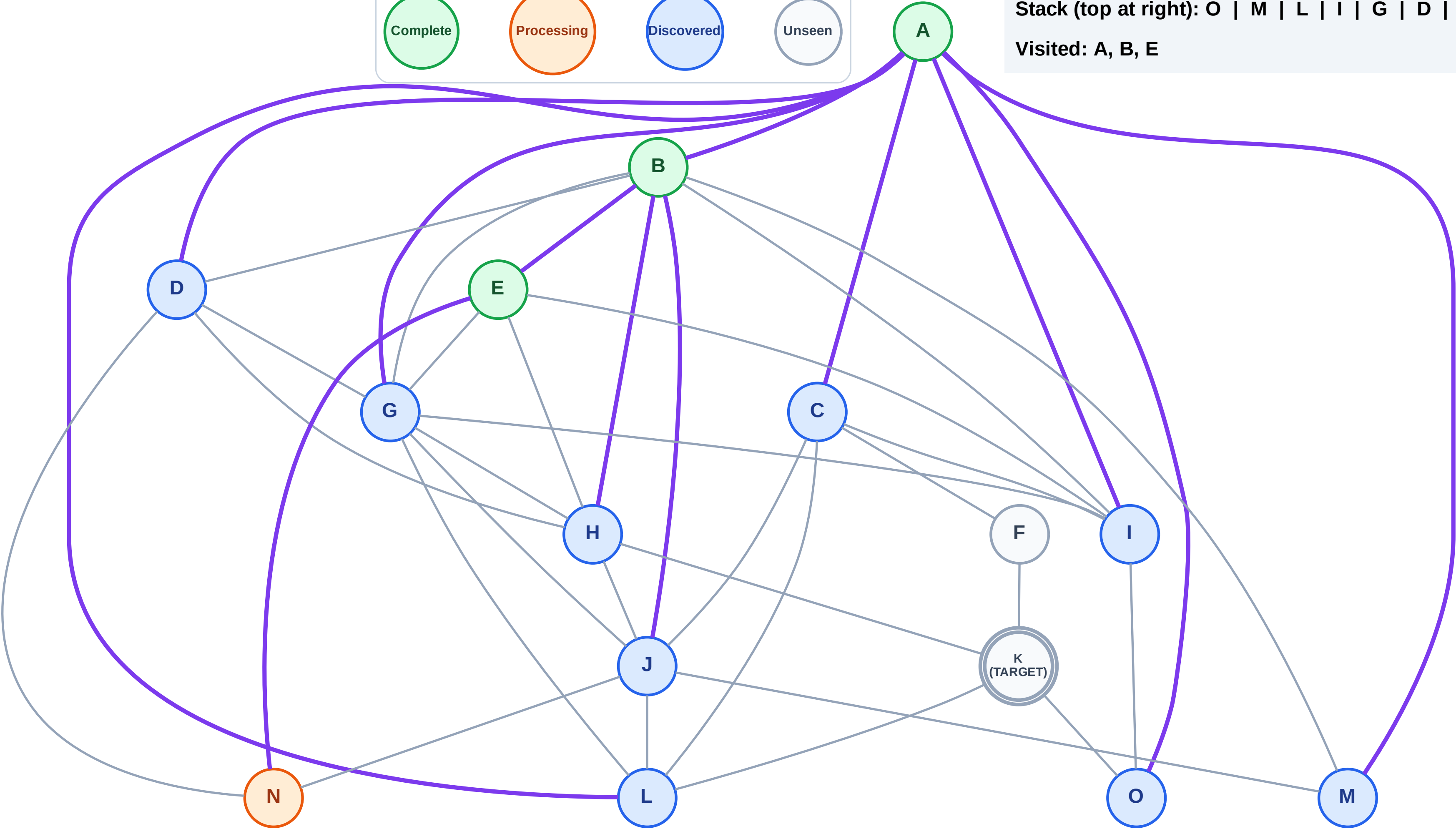
Processing

Discovered

Unseen

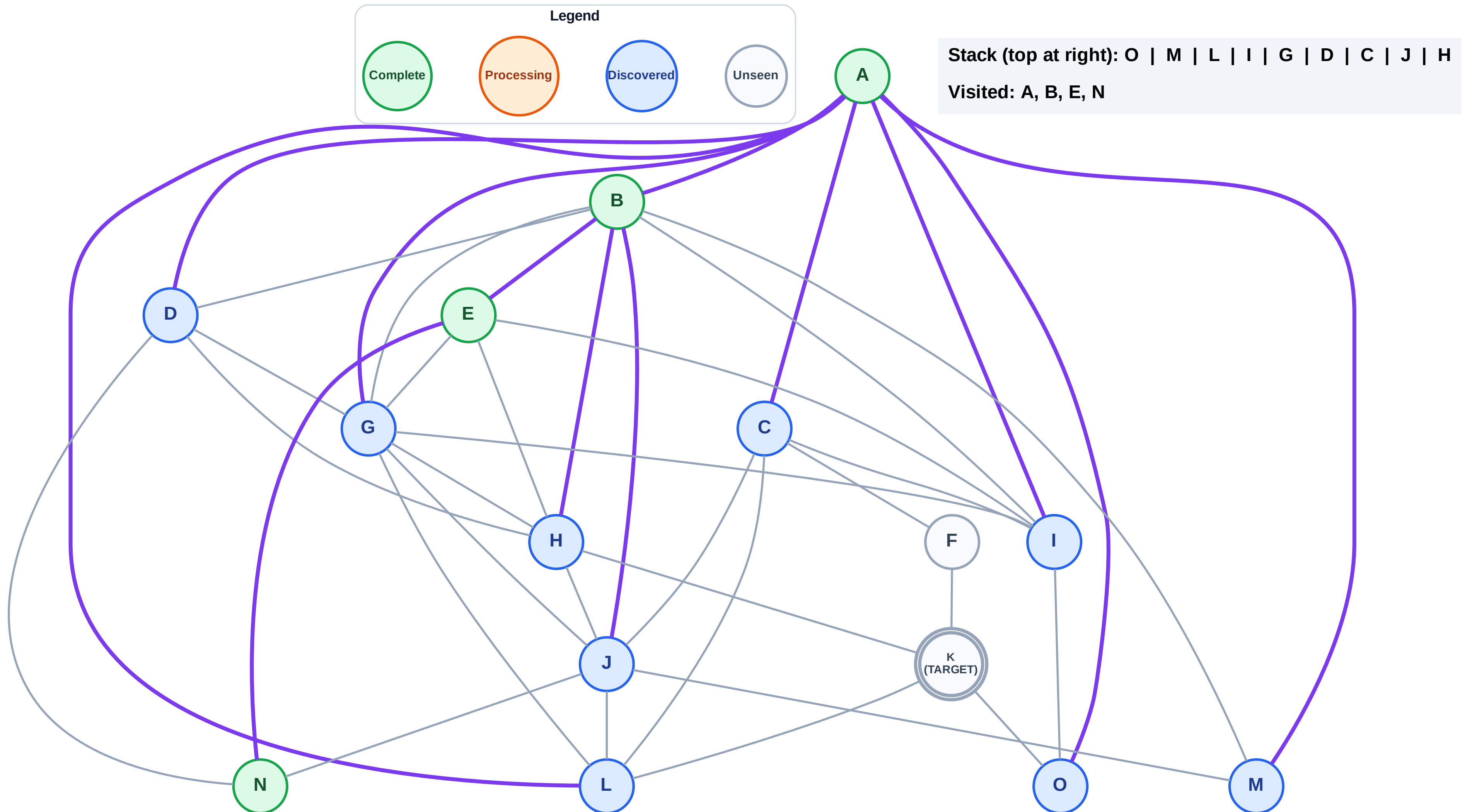
Stack (top at right): O | M | L | I | G | D | C | J | H

Visited: A, B, E



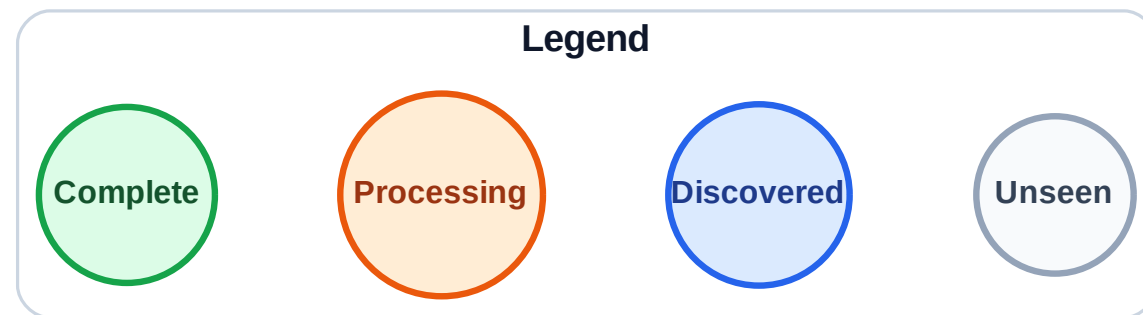
Step 9: No new neighbors from N

Step 9: No new neighbors from N



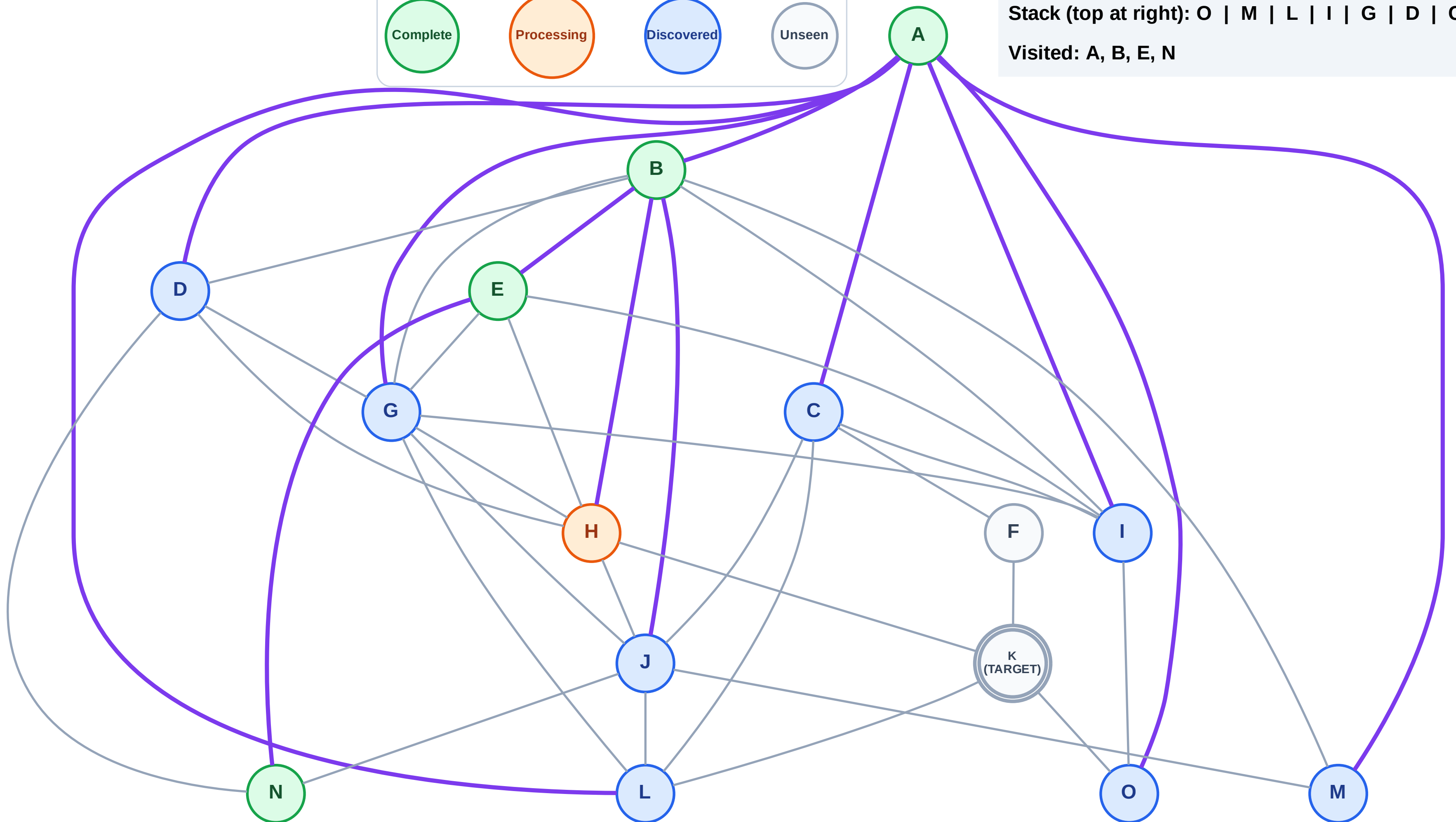
DFS Traversal

Step 10: Process node H



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, N



DFS Traversal

Step 11: Discover K from H

Legend



Complete



Processing



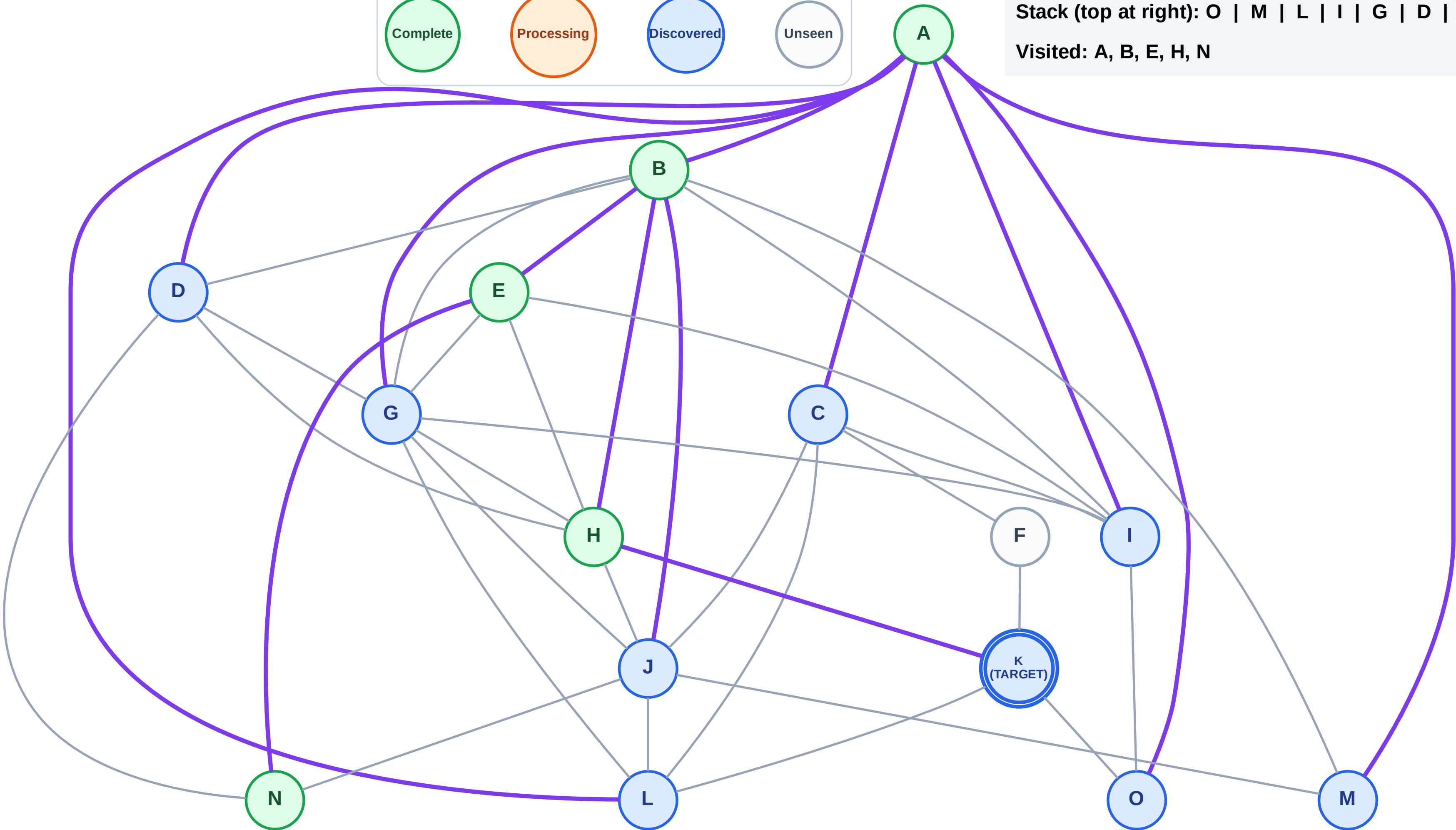
Discovered



Unseen

Stack (top at right): O | M | L | I | G | D | C | J | K

Visited: A, B, E, H, N



DFS Traversal

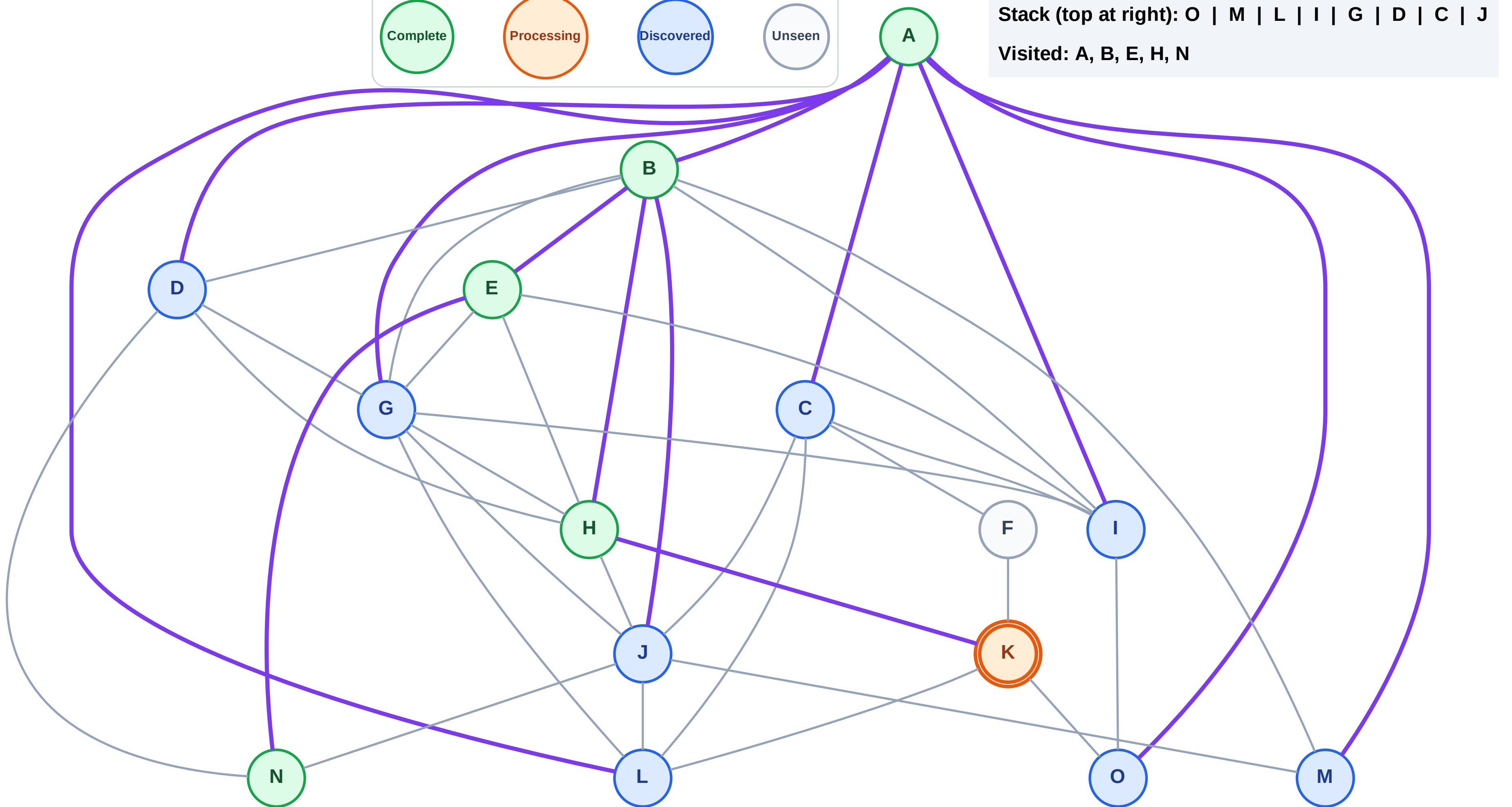
Step 12: Process node K

Legend



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, H, N



DFS Traversal

Step 13: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend



Stack (top at right): O | M | L | I | G | D | C | J

Visited: A, B, E, H, K, N

